13

Ultrasound propagation in foods and ambient gases: principles and applications

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13.1 Introduction

Ultrasound instrumentation is already established for a number of process control tasks in the food industry. These applications include the measurement of level, flow and suspended solids concentration and the detection of interfaces between liquids. The concentration of certain simple solutions is also being monitored with current instrumentation. Automatic qualitative distinction between food and cleaning liquids is carried out in other situations. One of the advantages of ultrasonic techniques is the option of non-invasive measurement using either clamp-on transducers mounted on existing pipework or transducers mounted on acoustic windows separating the transducer from the processed material. Whether this option is feasible in a particular application, depends on the acoustic impedance and attenuation of the processed material and the pipe or container (see Sections 13.4 and 13.5 and Table 13.1).

Recent developments in ultrasonic non-contact measurement even allow the characterization of materials at a stand-off distance where neither the processed material nor the container are contacted (see Section 13.2.5). Non-contacting NDT (Section 13.2.4) and the rapid scanning of large areas in flow measurement (pages 360, 391–2) have become possible with new transduction systems. The new measuring systems include laser ultrasonics, an air-coupled ultrasound transmitter used with a laser vibrometer for detection, a laser-EMAT transducer system and electrostatic transduction methods for measurement in both air and water. These methods will find their applications but the clamp-on or window-mounted transducers will continue to be widely used. Details on the established systems are given in Chapter 12.

In this complementary chapter, the underlying principles are described and illustrated with data for the relevant properties of foodstuffs, ambient gases and container materials. Techniques developed recently for a wider range of applications in the food industry, and awaiting commercial implementation, are discussed within the context of the principles involved. The new applications include, for example, the measurement of solid/liquid ratios for fats or of the original gravity of beer, or the determination of the average size and the size distribution of droplets in emulsions. Another interesting application is the

scanning of pipe contents showing a cross-sectional image of solids being carried by a liquid. The potential for a much wider range of applications with good reliability is based on advances in the understanding of the ultrasonic properties of foods and also in transducer technology and in electronics and mathematical methods for data acquisition and evaluation. Progress in these areas provides the basis for the technology transfer to the food industry of many measurement techniques that were initially restricted to the application in other sectors. An overview of ultrasound measurement applications for the food industry, either implemented or at the research stage, is presented in the Appendix.

Sections 13.3 to 13.5 are focused on the propagation of ultrasound in food liquids. The ultrasonic properties of ambient gases and solid foods are discussed more briefly. A few data for container and transducer materials are given in the tables in this chapter. The properties of the materials of processing vessels and pipes in the food industry are quite similar to those in other industries, and a description of these can be found in general ultrasonics engineering guides. (One container material unique to the food industry is discussed very briefly, namely egg shells.) The overview of the principles underlying ultrasonic techniques follows the theoretical treatment presented in the textbooks by Blitz (1963), Cracknell (1980), Crawford (1968) and Lynnworth (1989) or (mainly for Section 13.5.3) in recent articles. Measured values and food application details are from articles and handbooks as indicated in the tables, figures and text. Experimental studies carried out at the LFRA have also been taken into account.

13.2 Overview of ultrasound applications

13.2.1 What is measured?

Longitudinal sound waves consist of periodic displacements of the particles within the medium, setting up regions of compression and rarefaction propagating through the medium. There are many parallels to the propagation of mechanical waves on coupled bead-loaded springs (Section 13.3.1). However, for sound waves, the regions of compression and rarefaction are associated with temperatures above and below the equilibrium, respectively, which in turn influence the excess pressure within these regions. The thermodynamic (as well as the mechanical) properties of the medium thereby influence wave propagation.

Sound waves with frequencies above the audible range ($f \geq 20$ kHz) are called ultrasound waves. A number of variables characterizing the propagation of an ultrasound signal through a medium can be measured. These include:

- the transit time for the transmission of a pulse through the sample (yielding the speed of sound for a known sample depth or vice versa)
- the attenuation of a signal transmitted through the sample
- the reflected and transmitted proportions of a signal meeting an interface between the sample and a medium with known properties (yielding the acoustic impedance of the sample)
- the reflection coefficient at two angles (yielding the acoustic impedance and the speed of sound of the sample)
- the amplitude, phase and frequency of a scattered ultrasound signal at specified angles
- the propagation characteristics of a Lamb wave (plate wave) propagating in a plate- or sheet-type sample or in a container wall in contact with a monitored liquid
- acoustic emission generated by defects under load (yielding information on vessel degradation), by running machinery or by processes such as grinding, drying or extrusion.

In addition, such ultrasound measurements can be performed at several frequencies. This range of measurement types, together with the relationships of these measured variables with the physical and chemical properties of gases, liquids and solids, makes a wide range of applications possible.

New measurement techniques for the food industry are often the result of technology transfer from applications in other industrial sectors. Applications both within and outside the food industry are therefore briefly reviewed in this section. Later sections discuss the ultrasonic properties of materials relevant to the food industry in more detail and describe applications. Only those aspects of measurement techniques that are specific to food applications are considered in this chapter. An overview on general experimental methods in ultrasound measurement (with an emphasis on food applications) has been given by Povey and McClements (1988, pp. 233–241). They refer to Breazeale *et al.* (1981) and Papadakis (1976) for more general reviews of ultrasonic techniques in the laboratory and in industry, respectively. Detailed information on ultrasonic measurements for general process control applications can be found in Lynnworth (1989) or, in compressed and updated form, in Lynnworth (1994).

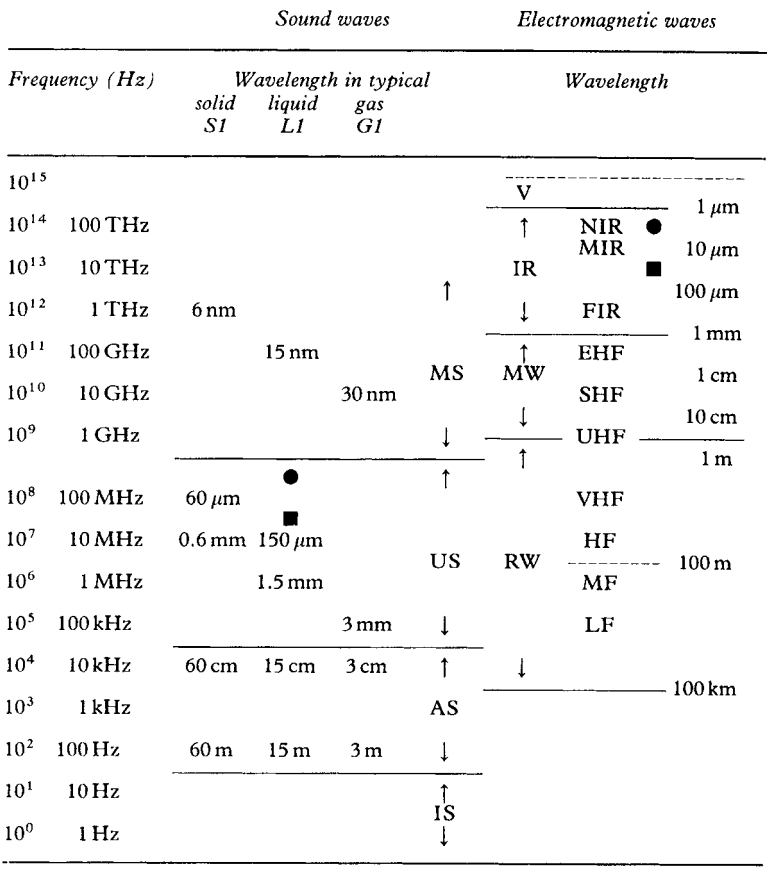
In consulting the presentations for general industrial applications, it is worth noting, however, that special considerations apply in the food industry. The scope of high-frequency transmission applications, for example, is restricted by the high attenuation of food materials including scatterers such as entrained bubbles or suspended particles (see Section 13.5.3). Pumped milk, for example, is highly aerated and, due to the resulting high attenuation for ultrasound, a careful choice of the applied method is necessary. On the other hand, this high attenuation caused by entrained scatterers can be used to determine the concentration of suspended solids or the stability of juices (see Section 13.5.3). Ultrasound techniques used in the food industry for the measurement of level, flow rate and suspended solids content are treated in Chapter 12.

13.2.2 Communication, detection and location

For a given target variable, it is often possible to choose between competing technologies, one using ultrasound waves and the other electromagnetic waves. The characteristics of their propagation and the implications for measurement applications are therefore compared here for the two wave types. In common with electromagnetic waves, sound waves can propagate through a gas, a liquid or a solid. Sound waves cannot propagate through a high vacuum, however. Their propagation characteristics, such as velocity or attenuation as a function of frequency, differ from those of electromagnetic waves, thus defining their respective application areas and associated frequency ranges.

Figure 13.1 shows the spectrum of sound waves in comparison with the electromagnetic spectrum. The audiosound range that we use for speech communication extends from about 15 Hz to 20 kHz, and wavelengths are in the metre and centimetre range. Similar wavelengths are encountered in radio, television and radar communication by freely propagating electromagnetic waves. This is the spectrum from the VHF radio range over the UHF range to the lower (by frequency) microwave range. The velocity of propagation in a gas is six orders of magnitude lower for sound than for electromagnetic waves. The frequency ranges for sound and radio communication are displaced correspondingly.

Moving up slightly in frequency and down to wavelengths of approximately 1 cm, we arrive in the detection and location ranges within the sound and the electromagnetic spectrum known as sonar and radar, respectively. Radar in the mid-microwave range is employed in the detection of aircraft. It can operate over long distances in air, but is



Acoustic media S1/L1/G1 have sound velocities 6000/1500/300 m/s^{-1}.
Examples for particulates (and distance) of dimensions corresponding to the wavelength are entered for illustration of scattering characteristics (Section 13.5.3) as follows:

■ suspended solids particle of 50 μm
● emulsion droplet of 5 μm

Sound wave ranges are: infrasound (IS), audiosound (AS), ultrasound (US) and microsound (MS).
Electromagnetic wave ranges are:

RW radiowaves, subdivided into earth-bound (MF, LF) and freely propagating (HF, VHF, UHF)
MW Microwaves
IR infrared, subdivided into far infrared (FIR), mid infrared (MIR), and near intrared (NIR; 0.7−2.5μm)
V visible (approximately 400–700 nm)
Ultraviolet, X-rays and gamma-rays are at higher frequencies and are not shown.

Fig. 13.1 Spectra for sound and electromagnetic waves.

strongly attenuated by water. The reverse is true for sonar, where sound waves instead of electromagnetic waves are emitted. Using a scanning system, the reflections from the object can be visually displayed showing bearing and range, similar to radar detection systems. Sonar was initially developed for the detection and location of submarines and later for the measurement of the depth of the seabed or of lakes. Sonar operates in the ultrasound range, that is with frequencies in excess of about 20 kHz, beyond the response limit of the human ear.

With ultrasound pulse-echo systems developed subsequently, the search for shoals of fish or the inspection of off-shore installations became possible. With a high-power echo

sounder it is even possible to detect a single fish of 35 cm length at a depth of 460 metres using 30 kHz (pulse length 500 μm, peak energy 8kW). With higher frequencies it becomes possible to identify the size and species of fish, but at the expense of operating over shorter distances only, since ultrasound attenuation by sea water increases with frequency (Cracknell 1980).

13.2.3 Level and flow rate measurement, ultrasonic imaging

The ultrasonic pulse-echo technique was later adapted to operate in other media, including gases and solids as well as liquids, and for a wide variety of applications, including level measurement and flaw detection. The tradeoff in the case of marine applications, namely better spatial resolution when the frequency is increased, at the expense of higher attenuation and thus shorter range, applies also to many of the industrial applications of ultrasound. However, the relationship between attenuation and frequency is often more complex in the media encountered here (see Sections 13.5.2 and 13.5.3), for example for transmission through gases and liquids carrying dispersed particles that scatter ultrasound.

It is also worth noting that resolution in ultrasound ranging is limited not only by long wavelengths but also by short pulse duration and low signal to noise ratios in the received echo (Cracknell 1980). Other factors influencing the choice of frequency are the transducer size and instrumentation costs. The ultrasound range extends to 500 MHz approximately. For process control measurements in the food industry, the frequency rarely exceeds 10 MHz.

The lowest part of the ultrasound range, up to 200 kHz approximately, is used for transmission through gases, for example in echo-ranging techniques for tank fill level measurement. In this frequency range, the ultrasound attenuation in most common gases is acceptably low so that a satisfactory reflected signal is received and the wavelength in gases is short enough to allow adequate spatial resolution. (Bats use much the same frequency range for their guidance.) In some applications of echo ranging through gases, microwave techniques are more appropriate than ultrasound. This is the case where strong temperature variations along the path of the beam may occur, for example in the ullage space of ships' holds.

Liquids attenuate ultrasound less than gases, in general, so that higher frequencies can be employed. For process control applications based on ultrasound echo ranging in liquids, frequencies in the range 1 to 5 MHz are commonly used. These frequencies are higher than for sonar since transmission over long distances (as would be encountered in the location of fish shoals) is not required and demands on spatial resolution or on accuracy in velocity measurement are more stringent. The velocity of sound is higher in liquids than in gases so that the wavelengths encountered in echo ranging through liquids are of the same order as for gases, despite the frequency difference. Details on ultrasonic level and flow measurement in the food industry are provided in Chapter 12. Additional notes are given here on pages 391–2.

The non-invasive measurement of flow rates and flow rate distributions (CPPM 1989, Rickey *et al*. 1991) is routinely used in the clinical sector. More recent developments include three-dimensional flow rate imaging (pages 391–2). Ultrasonic imaging at 12 MHz has found well-established clinical applications (Crecraft 1983; Wells 1977; Newhouse 1988). Ultrasound is also used for the assessment of back fat in live agricultural animals (Section 13.3.5). For three-dimensional real-time scanning, special techniques allowing the implementation with a reduced (and therefore more practical)

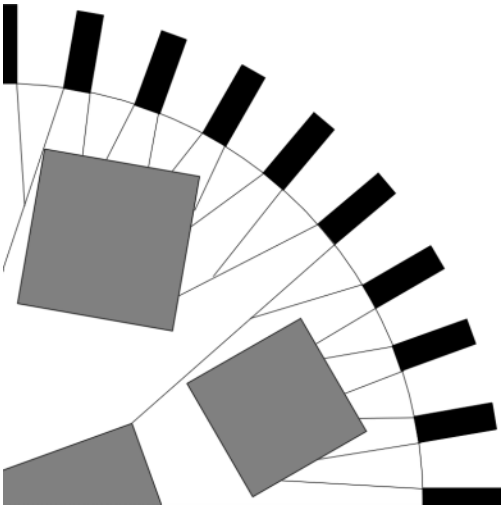


Fig. 13.2 Ultrasound tomography for industrial processes. Schematic diagram showing part of a pipe cross-section for a liquid carrying solid items illuminated by ultrasound beams. 36 transducers are mounted around the pipe in the system developed by Schlaberg *et al*. (1998).

number of transducer channels have been developed (Nikolov and Jensen 2000). Fuzzy logic has been deployed for improved ultrasound image generation (Hall *et al*. 2000).

Ultrasonic scanners designed for clinical applications are, in general, unsuitable for industrial applications because industrial processes can present a mixture of strongly inhomogeneous materials with great variations of acoustic impedance. Most of the signal is then reflected at the first interface and the objects further down are not imaged. To overcome this problem, several sensors can be mounted around the area of interest so as to obtain views from different angles. On this basis, an ultrasound tomography system for industrial processes has been developed. Schlaberg *et al*. (1998) constructed a circular array of 36 transducers mounted around a pipe cross-section (Fig. 13.2). It generates 30 frames per second with images of 100 $\times$ 100 pixels.

13.2.4 Non-destructive testing and non-destructive evaluation of microstructure (NDT and NDE)

In non-destructive testing (NDT), the ultrasonic flaw detector is well established (see for example, Krautkrämer and Krautkrämer 1983, American Society of NDT 1991) with frequencies up to about 20 MHz for the examination of fine-grained metals and thin objects. Ultrasonic thickness gauges are used in the inspection of components such as turbine blades; a resolution of 0.15 mm can be achieved with a frequency of 20 MHz.

Ultrasonic wavelengths in water and in many liquid and semi-solid foods at 5 MHz are similar to those in steel at 20 MHz, so that one could expect a similar resolution. However, in high-speed processing lines it is difficult to arrange for good acoustic contact between the transducer and a food coating layer, for example, without disturbing or damaging the layer. In laboratory studies, immersion tanks have been employed to measure the thickness of egg shells, and the use of soft-tipped probes has been proposed as an alternative (see Section 13.3.5).

Non-contacting NDT methods developed more recently offer more potential for applications in processing lines. Dixon *et al*. (1999) use a pulsed laser to illuminate the

sample together with an electromagnetic acoustic transducer (EMAT; see Frost 1988) to measure the ultrasonic modes generated in the sample by the laser pulses. Using this method, they scan the sample for sub-surface micro-defects.

As another potential problem in food applications, resolution can be limited by weak echoes as they result from high attenuation in many foods. Progress in signal evaluation techniques helps with this type of problem. The interpretation of pulse-echo measurements in ultrasound NDE can be facilitated by a signal evaluation technique that deconvolutes the distortion resulting from the transducer characteristics (Olofsson and Stepinski 1999).

Polycrystalline solids scatter sound in a manner similar to gases or liquids containing fine-grained particles. In the determination of metal grain size, scattering techniques using frequency scanning are now gaining ground despite the need for considerably more signal processing compared with the older single-frequency attenuation method. For example, scanning from 3 to 12 MHz is used to determine grain sizes in the range of 50 to 350 m in steel (Lynnworth 1989). For the food industry, methods for applications such as the determination of droplet size in emulsions, for example, where several scattering mechanisms overlap (Section 13.5.3) are of interest.

NDT and NDE are widely applied not only to metal inspection but also to the investigation of fibre-reinforced composites for microstructural parameters (Cai and Williams 1999) and to the examination of tissue structure. While the latter type of examination is applied primarily in the medical area, applications in the assessment of food texture are worth investigating. When an ultrasonic wave travels through biological tissue, the interaction of the wave with the inhomogeneities in the medium leads to absorption and scattering of the incident wave. Information on the size and spacing of the scattering structures may be extracted in the frequency domain via analysis of the power spectrum or in the spatial domain from the autocorrelation function. Scatterer spacings in the range 1 mm to 20 mm have been determined in this manner (Edee 2000) to characterize tissue structure.

Ultrasound scattering from variations of density and compressibility has been studied for a tissue-mimicking material (glass beads in agar) by Esward and Blakey (1998). They determined scattering cross-sections for scatterers with sizes in the mm range using ultrasound at about 2 MHz. The tortuosity of bone microstructure has been studied by Nicholson and Strelitzki (1999). This study refers to earlier investigations of air-saturated porous foams.

Scanning acoustic microscopes (SAM) were first applied to stress measurements in amorphous metals then the application was extended to polycrystalline metals (Okade and Kawashima 1998). An acoustic microscope for materials with high attenuation such as bone (but also relevant for other air-saturated porous materials) has been presented by Petersen *et al*. (1998). This instrument operates at relatively low frequencies around 10 MHz (compared with about 100 MHz for the earlier acoustic microscopes) yet provides high resolution.

13.2.5 Concentration measurement and classification of materials

Current ultrasound applications in the food industry are primarily level and flow measurement, but also the monitoring of suspended solids concentration and the detection of interfaces between immiscible liquids (see Chapter 12). In other industries, primarily the petrochemical and nuclear sectors, the relationship between the acoustic properties of a material and its chemical composition is being used more widely to determine solute

concentration or to identify liquids. The possibility of mounting ultrasound transducers non-invasively on the outside of vessels containing hazardous materials is particularly suitable for these industries (Asher 1983). A compact immersible sensor for hostile environments has been developed by Greenwood *et al*. (1999). With this sensor, the transducers are not in contact with the monitored liquid but are separated from the liquid by a plastic wedge.

The possibility of non-contact application is also attractive to the hygiene-conscious food industry. Concentration measurements are already used for the monitoring of simple solutions, such as sodium hydroxide cleaning solutions, or pure sucrose or sodium chloride solutions, and in the control of coffee extract dilution and corn starch processing.

The idea of an ultrasonic 'solution analyser' (better termed a concentration meter) for alcoholic beverages and dairy liquids was proposed in the late 1940s. Interest in practical applications in the food industry received a new impetus in the late 1960s, when the implementation of ultrasonic concentration monitoring was successful in the chemical industry, where ultrasound applications soon expanded to other tasks (Zaccharias and Parnell 1972; Steele 1974). (The term 'moisture measurement' was sometimes used to describe the determination of the water content of aqueous solutions.) Applications in the in-line monitoring of food properties (rather than just process variables such as flow rate or fill level) have so far largely been restricted to a few simple process streams or else to a distinction of beverage from cleaning solution. Recent advances in the understanding of the ultrasound properties of food and the greater feasibility of multivariable measurement in line is now offering the potential of widening the range of applications in the food industry. The measurement of solute concentration or the water content of solutions is discussed in more detail in Section 13.3.4.

While most ultrasonic measurements for the determination of solute concentration and solid/liquid fat ratio in food materials (Section 13.3.4) would rely on the measurement of ultrasound velocity, attenuation can be used to determine the concentration of the solids content of slurries (Section 13.5.3) or the admixture of other gases to CO_2 (Section 13.5.2).

In the submersible sensor by Greenwood *et al*. (1999), the reflection coefficient at two angles is measured to determine the density and the speed of sound of the liquid in a pipe or tank. This is achieved by mounting six transducers around the arc of a semicircular plastic wedge. Only the straight side of the wedge is in contact with the liquid. The sensor has been designed not only for radioactive materials but also for process control applications in the petrochemical, chemical and food processing industries. Since only the plastic wedge is exposed, the device is suitable for hostile environments which should also include CIP (cleaning in place) procedures in the food industry with hot caustic solutions. The device has been tested in sugar solutions and in slurries. Another method for hostile environments is the separation of the sensor from the process by a buffer waveguide developed for applications in very hot fluids (Liu *et al*. 1998). Another immersible sensor for the characterization of liquids has been developed by Scandelari *et al*. (1998). This is a truly compact device with a thickness of 0.2 mm and a width of less than 1 cm.

A number of ultrasonic methods have been developed for non-contact concentration determination and classification. One of these systems operates at a stand-off distance of three metres from the vessel to classify the liquid within (Kaduchak *et al*. 2000). This is based on an air-coupled ultrasound transmitter together with a laser vibrometer for measuring the container response. The system measures the propagation of a Lamb wave

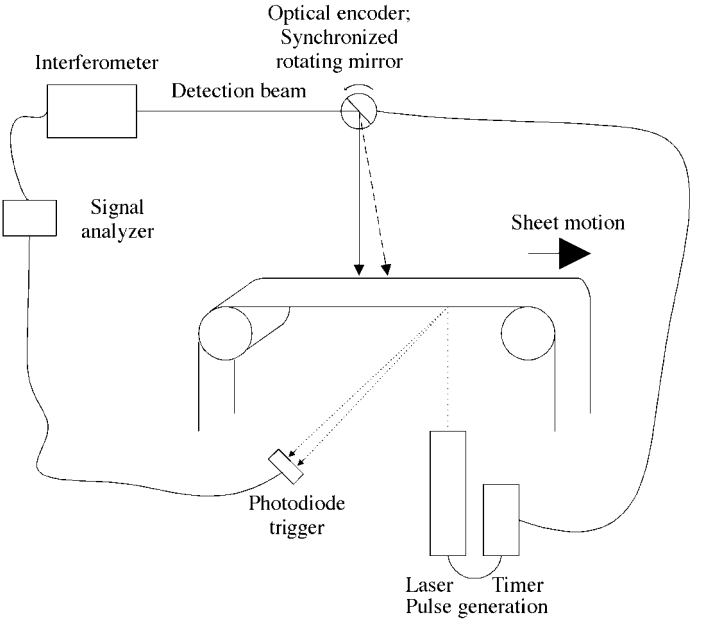


Fig. 13.3 Measuring the ultrasonic wave velocity in a moving sheet material from a stand-off distance. In this laser ultrasonics system by Ridgway *et al*. (1999), the laser pulse generates ultrasonic waves in the sheet material. The wave propagates along the sheet and is measured at a known distance away using a non-contact interferometric technique.

(plate wave, see Section 13.3.2) and measures the properties of the liquid in the vicinity of the container wall.

Alternatively, laser ultrasonics (Scruby and Drain 1990) can be employed as a technique for the measurement of acoustic properties at a distance. The sensing system has no contact with either the monitored liquid or its container nor, in the case of a solid sheet material, with either material or the belt transporting it. In this system, a laser pulse with focus in the monitored processed material generates an ultrasonic wave by thermal expansion or an ablation shock wave. This ultrasonic wave propagates in the material and is detected at a known distance using a non-contact interferometric technique (Fig. 13.3). This has been tested in the non-contact monitoring of the elastic properties (Young's modulus) of paper moving on a belt at 12 m/s (Ridgway *et al*. 1999). A potential drawback of this method is the localized increase in temperature which might affect delicate materials.

13.2.6 Passive ultrasound equipment: acoustic emission monitoring

The ultrasonics chapters in this book describe measurements of food properties or of process variables, based on the interaction of an applied low-intensity ultrasound signal with the food, ambient gas and container material. Before describing these in detail, it is appropriate to include an alternative acoustic measurement technique in this overview section. In this technique, the equipment does not emit an acoustic signal but analyses audiosound and ultrasound created by the process itself. This passive technique is known as acoustic emission monitoring (AEM; see Miller and McIntire 1987). AEM operates using wide-band transducers in the ultrasonic frequency range from 100 kHz to 1 MHz

(Hill *et al*. 1998). It has been used commercially since the late 1960s for the testing of pressurized systems in the chemical and aerospace industries. Routine applications include quality assurance of new vessels, regular evaluations for vessel degradation and continuous monitoring of processes. Even quantitative monitoring is feasible with the help of a simulated AE source (Evans *et al*. 2000).

The AEM equipment listens to leaks or bubbles, mixing or grinding operations, solids flow or bearings. It detects deviations from the standard pattern, indicating a defect in a component or an out-of-specification processing operation. It can indicate the end point of an operation such as milling. The sound (in the audiosound or ultrasound range) that is detected in acoustic emission can result from stresses or impact created within the process that is monitored; or stresses can be induced to generate sound emission as a crack grows in the processed material or the vessel, for example. These techniques have evolved from the ancient methods of tapping objects and listening to deduce unseen properties such as the soundness of crocks or the ripeness of water melons. Applications are primarily in detecting flaws and anticipating failure in engineering structures (see for example Szilard 1982; and issues of the *Journal of Acoustic Emission*, published by the Acoustic Emission Group, University of California).

An on-line monitoring system with process-integrated measurement of acoustic emission to evaluate grinding processes has been developed by Tönshoff *et al*. (2000). Another application of acoustic measurement with potential for food applications is the monitoring of foam evolution and foam stability (Rzeszotarska *et al*. 1998). Although an early application of AEM to the assessment of the quality of sulphur samples was reported in the *Codex Germanicus* in AD 1350, the extent of AE from chemical reactions was realized much later (Betteridge *et al.* 1981). Now, the monitoring of hydration, titration, precipitation and electrolysis processes by AEM is being investigated, or an enzyme is immobilized on an AE sensor which then monitors effervescence from gas production (Wade *et al.* 1991).

Drying processes in biological tissues can cause AE. Under conditions of draught, suction pressures in plant conduits can reach −10 MPa. When these tensions are released by cavitation, AE occurs. This is being investigated for irrigation control. The drying of woody tissues can also be monitored to control drying rates. One can conjecture that it may be possible to develop AEM techniques for the observation of drying processes in foods with a fibrous structure or for the assessment of other water migration processes in foods.

Applications in the monitoring of food processing operations have been developed at the LFRA. These include the monitoring of food powder flow rates and of the impact of cleaning fluid sprays on vessel surfaces. An acoustic emission sensor mounted underneath a cooker extruder provided a signal related to the density of the final food product emerging from the extruder (Roberts and Wiltshire 1990a).

'AE can be simple to detect and difficult to interpret' (Wade *et al.* 1991). Neural network techniques (Jansson 1991) have been explored early on to help with the analysis of AE spectra (Zgonc and Grabec 1990). This technique has now reached the point where AE sources within a material can be localized (Grabec *et al*. 1998a, Prevorovsky *et al*. 1998). The application of a sensory-neural network for the AE monitoring of manufacturing processes has been described by Grabec *et al*. (1998b).

13.2.7 High-power ultrasound equipment

Instrumentation for ultrasonic measurements in industrial process control uses low signal intensities, or short pulses of higher peak intensity separated by long 'dead' times. This

low time-averaged signal intensity leaves the sample unchanged. The sample is also unaffected by AEM, where the applied ultrasound intensity is zero. At the other end of the power scale is the ultrasound equipment designed to take part in the food processing operation rather than monitor it. Many readers will have used ultrasonic high-power appliances in sample preparation in the laboratory; others already use ultrasonic food processing plant or are considering this option. The effects of high-power ultrasound are therefore briefly introduced in this section, although they do not occur with the instrumentation used in ultrasonic measurement.

High continuous power levels are used in equipment for sonoprocessing, such as ultrasonic cleaning baths, homogenizers, emulsifiers or (microbial) cell disrupters (see Ensminger 1973; Puskar 1982 for a general introduction to high-power ultrasonics). The effect of these high ultrasound power appliances on the medium is often due to cavitation, that is the sudden collapse of gas bubbles in the liquid and the associated high pressures. It is caused by rapidly alternating pressures of high amplitude. Frequencies of typically 2050 kHz are used since the threshold intensity for cavitation rises rapidly for higher frequencies, and lower frequencies would be within the audio range (Cracknell 1980; Young 1989, Ciuti *et al*. 1998, Laborde *et al*. 1998). High-power ultrasound can also induce or promote chemical reactions, and this application has acquired the name sonochemistry (Henglein 1987; Heusinger 1987; Lorimer and Mason 1988, Jossinet *et al*. 1998). The reaction rate is higher owing to the high temperatures, particularly in the cavitation regions, and to the increased intermolecular contact in regions of high and fluctuating pressures.

Sajas and Gorbatow (1978) have studied the nature of the changes in meat and meat products during high-power ultrasound treatment and have recommended treatment procedures based on this. This work has included the following meat-processing technologies: tenderizing of muscles, curing of meat, emulsifying sausage mix for scalded sausages, preparing emulsions from spice extracts and spray drying of meat products. They have carried out a detailed investigation of the effect of the properties of ultrasound treated meat and meat products, including the physical structure (the fibrous structure of meat and the droplet size distribution in emulsions), the biochemical properties (fatty acids composition, amino acids composition and the presence of enzymes), the nutritional properties (digestibility) and the microbial flora.

Many other high-power ultrasound applications exist in food processing (Roberts and Wiltshire 1990b). Sonoprocessing increases extraction rates, promotes crystallization with the formation of finer crystals and homogenization resulting in stable emulsions with fine droplets, thins polysaccharide solutions and helps to degas liquids and clean contaminated surfaces. Cavitation occurs preferentially in existing fine bubbles or small voids at liquid-solid interfaces (owing to imperfect wetting) and in high-temperature regions. This can be used to concentrate the applied energy at a contaminated surface or at a heat exchanger surface, for example. Standing wave techniques are used in separation processes.

Sonochemistry enhances oxidation processes (helpful in the accelerated ageing of alcoholic beverages, undesirable in the processing of fats). It promotes catalytic reaction by keeping catalyst surfaces clean and by increasing the area of interfaces between immiscible liquids and gases taking part in the reaction (for example the hydrogenation of oils). One high-power technique not relying on cavitation is ultrasound-assisted drying. This is based on resonant absorption of bubbles (see Section 13.3.4) in capillaries and resulting pressure differentials and on increased convection induced by the passage of high-intensity sound waves in air. None of these effects occurs at the time-averaged power intensities used in ultrasound measurement.

13.3 Speed of sound

13.3.1 Velocity of propagation for bulk longitudinal waves

For a summary of the symbols used in this chapter, see p. xxvii. The speed of sound is

$$c = (E_M/\rho)^{1/2}$$

where, for longitudinal (compressional) waves, the elastic modulus E_M is taken as

$$E_M = K_s = \gamma P \quad \text{for gases}$$

$$E_M = K_s \quad \text{for liquids}$$

$$E_M = Y + \tfrac{3}{4}G \quad \text{for bulk solids}$$

The three elastic moduli are given by the ratio of applied stress to resulting strain:

$K =$ (hydrostatic pressure)/(relative decrease in volume)

$Y =$ (stretching tension per cross-sectional area)/(relative increase in length)

$G =$ (shearing force per area)/(angle of shear)

They are a measure of the compressibility ($1/K$), the stretchability ($1/Y$) and the rigidity (G) of a material. A similar, more familiar waveform is the mechanical wave propagating on coupled springs (see Fig. 13.4).

13.3.2 Other ultrasonic wave types

In solids, transverse (shear) ultrasonic waves, as well as longitudinal (compressional) waves, will propagate. Here the displacement is transverse to the direction of wave propagation rather than in the same direction. Shear waves can be induced by applying a transverse wave transducer or by mode conversion of longitudinal waves at an interface. The velocity of transverse waves is

$$c_{\text{shear}} = (G/\rho)^{1/2}$$

For animal bone, the shear wave velocity has been measured as 1970 m/s (Kaye and Laby 1986). Very few foods have such high rigidity, and shear wave velocities in food are therefore much lower.

Transverse (or shear) ultrasonic waves are not supported by gases or by most liquids. They are strongly attenuated even in viscoelastic liquids. In food gels (gelatine, pectin, agarose or sugar and gelatine in water), attenuation for shear waves was found to be too high for transmission through the sample. The shear velocity (0.5 m/s and below) was by three orders of magnitude lower than the compressional wave velocity and thus very

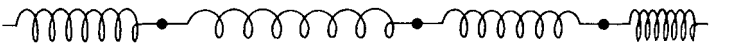


Fig. 13.4 Coupled bead-loaded springs. Regions of compression and stretching for a longitudinal wave carried by coupled springs loaded with beads. At equilibrium, the distance between the beads is a; the average mass per length is $\rho_0 = M/a$; the spring constant is K_{sp}. The wave propagates with velocity $c_p = K_{sp}a/\rho_0)^{1/2}$. The form of this equation is similar to that for the propagation of sound waves.

difficult to measure. Even for an exceptionally stiff food gel (40 per cent gum arabic in water), the shear wave velocity (2.6 m/s) is quite small compared with the compressional wave velocity (Povey 1989).

Low velocities combine with high attenuation to render shear wave measurements in foods extremely difficult. For solid foods, where the intrinsic shear wave properties could be expected to be more favourable, other factors conspire against ultrasound transmission (scattering, coupling problems, etc.; see Section 13.3.5). The application of very-low-frequency shear waves in the audio range has been suggested but here the sample geometry can be a significant factor. It is possible that transverse waves in foods could play a role in acoustic impedance techniques (Section 13.4.3) as these sample the interface properties and can be applied to highly attenuative acoustic media. However, no food applications that bear this out conclusively are known to the author.

Readers of the ultrasonics literature are likely to come across Rayleigh waves. These are surface waves and are among the principal wave types encountered in industrial measurements. Rayleigh waves have both a longitudinal and a transverse component and propagate in solids only. For food industry measurements, Rayleigh wave propagation in container materials and attached ultrasound waveguides can be relevant. Rayleigh waves can be generated either by a special transducer or by mode conversion of bulk waves. They are characterized by a very low attenuation compared with bulk waves, and this is used in some sensing systems to transmit the signal for remote measurement applications in vessels where access for transducer maintenance is not possible. ('Remote' sensing in industrial applications usually implies distances of tens of metres, rather less than in the remote sensing from satellites.)

Lamb waves are also restricted to propagation in solids. Lamb showed that a solid plate can resonate at an infinite number of frequencies. Laminar defects just below the surface can be detected by observing the resonances in the plate between surface and lamination (Blitz 1963). Lynnworth (1989) describes the edge view of the resonating plate as resembling 'a straight snake swallowing a series of equally spaced eggs' (for symmetrical Lamb waves) or as 'a flag in a steady wind, with periodic waviness' (in the asymmetrical case). (See also Lynnworth 1989, Tables 3.1, 3.2 and 3.5 on ultrasound wave types.)

The leakage of Lamb waves in transducer or vessel walls into surrounding liquid can be used to detect the presence or absence of the liquid at a particular level. Level switches and level meters using this principle are described in Chapter 12, Sections 12.3.2 and 12.3.3. Measurements based on Lamb wave propagation can also be used to classify a liquid layer in contact with a container wall illuminated with ultrasonic waves (Kaduchak *et al*. 2000) or a liquid layer coating an immersed Lamb wave device (Scandelari *et al*. 1998). Lamb waves can also propagate in a sheet material and this is the basis of a system for the monitoring of the elastic properties of paper moving on a belt (Ridgway *et al*. 1999).

Lamb wave devices belong to a group of special acoustic devices, the surface acoustic wave (SAW) devices, which are used in ultrasonic high-frequency applications. The propagation of the ultrasonic wave at the surface of the device is influenced by changes in the physical properties of the surface layer. Rather than just measuring the acoustic impedance and velocity of the sample liquid in contact with the device, higher specificity can be obtained by coating the sensing surface with specifically absorbing or binding layers. A wide range of sensors for chemical and biochemical variables has been developed on this basis (see Chapters 20–23 for chemical sensors and biosensors). A review on acoustic wave sensors has been presented by Vellekoop (1998).

Following these brief notes on transverse waves and on surface waves, the remainder of the chapter is concerned with bulk longitudinal ultrasonic waves.

13.3.3 Speed of sound in gases

In a gas, only longitudinal sound waves will persist, as shear stresses are not supported. The speed of sound is

$$c = (\gamma P/\rho)^{1/2}$$

For an ideal gas,

$$c = (\gamma RT/M)^{1/2}$$

The speed of sound for the ideal gas is thus pressure independent. This is not the case for non-ideal gases, for example organic vapours or gases under high pressures, where

$$c = [\gamma RT + 2BP)/M]^{1/2}$$

In both cases, the speed of sound increases with temperature and decreases for increasing molecular weight. The ratio γ of the principal specific heats is approximately 1.4 for diatomic gases such as nitrogen or oxygen, and 1.3 for carbon dioxide.

For the propagation of ultrasound in certain polyatomic gases such as carbon dioxide, velocity dispersion can occur (that is the speed of sound varies with the frequency), still within the frequency range for practical applications in gases. This velocity dispersion will be accompanied by strong attenuation, often precluding measurements (see Section 13.5.2) in such gases. As a consequence, the frequency dependence or the additional pressure and temperature dependence of the dispersive velocity will not usually be a concern in food industry applications.

13.3.4 Speed of sound in liquids

Influence of solutes, solvents and temperature

In most liquids, in common with gases, only longitudinal waves will propagate. (As an exception, shear waves can be observed in highly viscous liquids when viscoelasticity occurs.) The speed of sound is

$$c = (K_s/\rho)^{1/2} = [c_p(\gamma - 1)/(\beta_p^2 T)]^{1/2}$$

For transmission through liquids, as in the case of gases, compensation for temperature variations will normally be required for any instruments relying on a measurement of the speed of sound (unless temperature is the target variable).

Whereas for gases the velocity increases with temperature, the reverse is true for most pure liquids (away from the critical temperature). For distilled water, however, the speed of sound exhibits a maximum at about 74 °C (see for example Kaye and Laby 1986 for further data). The temperature dependence changes with the introduction of solutes or of miscible liquids. This is of interest when compensating measured velocities for temperature changes. For aqueous sugar solutions, for example, the shape of the curve for the temperature dependence of the speed of sound will vary considerably with the concentration and temperature range of the solution (see Fig. 13.5).

Solute molecules influence the speed of sound by acting on the solvent as structure formers or structure breakers, and also by influencing the density of the solution. For sugar solutions, the increase of the sound velocity with concentration (Figs 13.5b, 13.6, 13.7) is well documented. The density increase accompanying the solute increase would

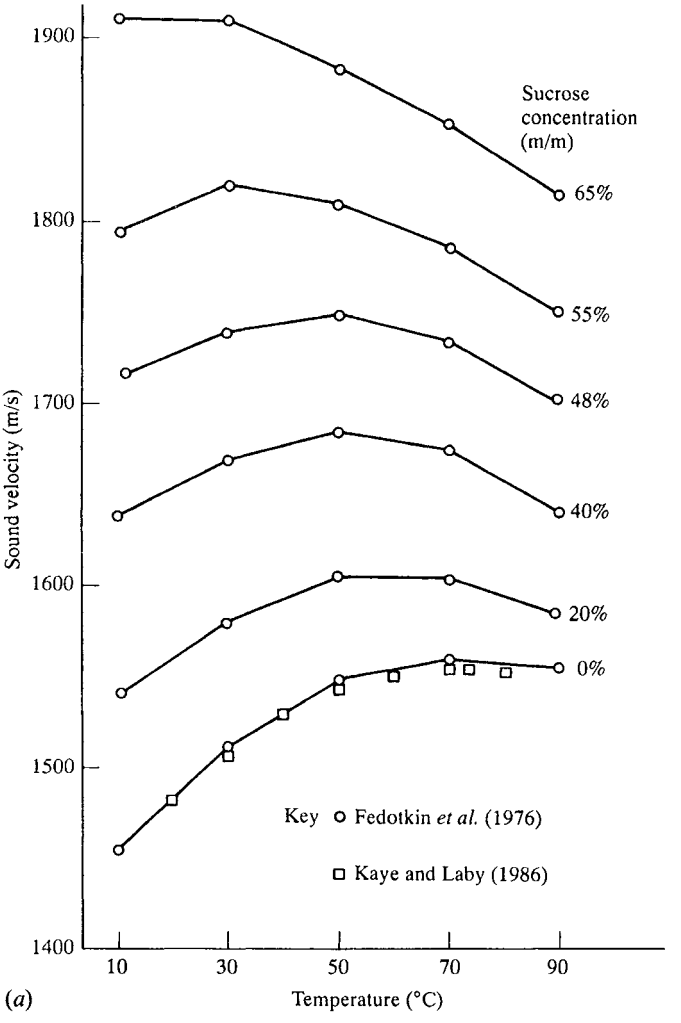


Fig. 13.5 (a) Sound velocity as a function of temperature for water and aqueous sucrose solutions. (b) Sound velocity as a function of sucrose concentration in aqueous solutions at 30°C. (c) Change of relationship between sound velocity and sucrose concentration with temperature. Data in Figure 13.5b from Schaafs (1963) (given there in tabular form), and in Figures 13.5a and c from Fedotkin *et al*. (1976) (as part of a graph there). To illustrate the error in rereading data from a small graph, data for water from Kaye and Laby (1986) have been entered in Figure 13.5a for comparison.

be expected to lead to a decrease in the velocity, and it is therefore clear that the velocity change is, instead, dominated by a decrease in compressibility with increasing solute concentration. This has been examined in more detail for monosaccharides by Smith and Winder (1983). The increase of sound velocity with concentration for aqueous NaCl solutions is shown in Fig. 13.8. As for the sugar solutions, the change of velocity with concentration is dominated by the change in compressibility rather than that in density.

Even within the group of monosaccharides, the speed of sound varies (at the same concentration), and there are even greater differences between monosaccharide and disaccharide or polysaccharide solutions (see Figs 13.6 and 13.7). This would pose a

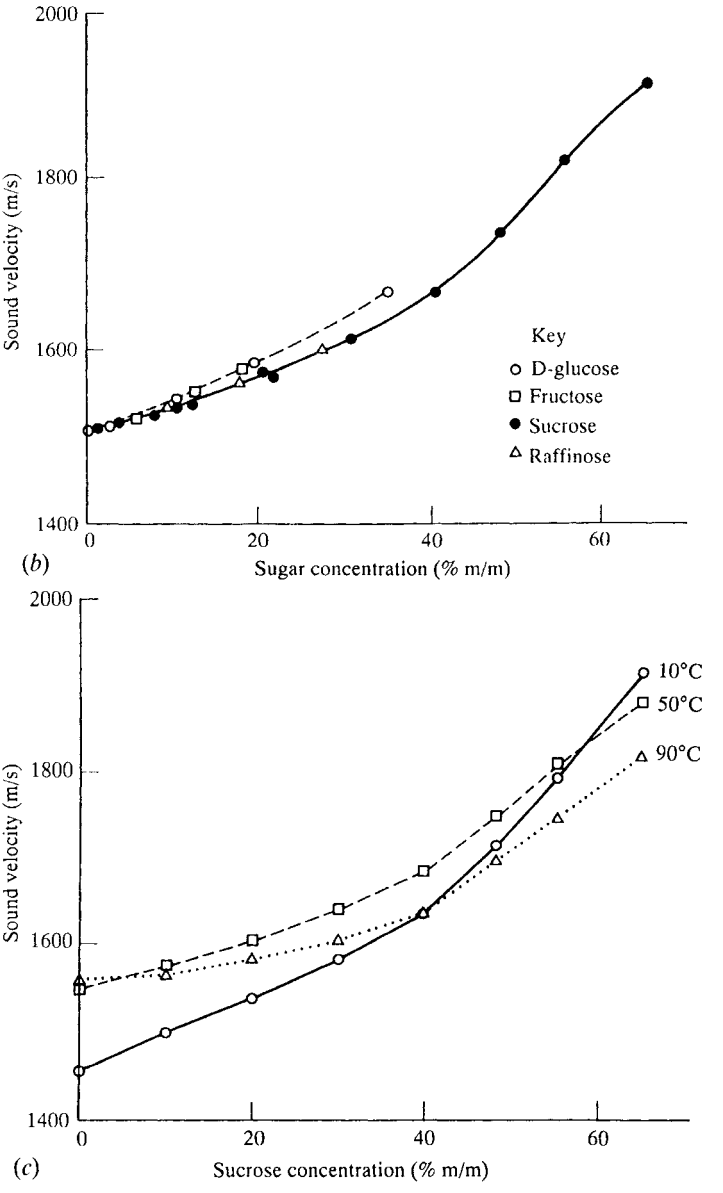


Figure 13.5 (contd.)

problem, for example, in the monitoring of the concentration of sugar syrups of varying saccharide composition by measuring their speed of sound. Similarly, it would introduce an error in monitoring the temperature of such a syrup (at constant concentration) by the same measurement. However, where the composition of the dissolved saccharides is constant, ultrasonic measurement is an option for concentration monitoring. An ultrasonic sensor for the in-line measurement of dissolved sugar concentration and for the characterization of slurries has been developed by Greenwood *et al.* (1999). The sensor is compact and immersible. It can be mounted in pipelines and containers. To assess homogeneity, several of the sensors can be distributed within a container.

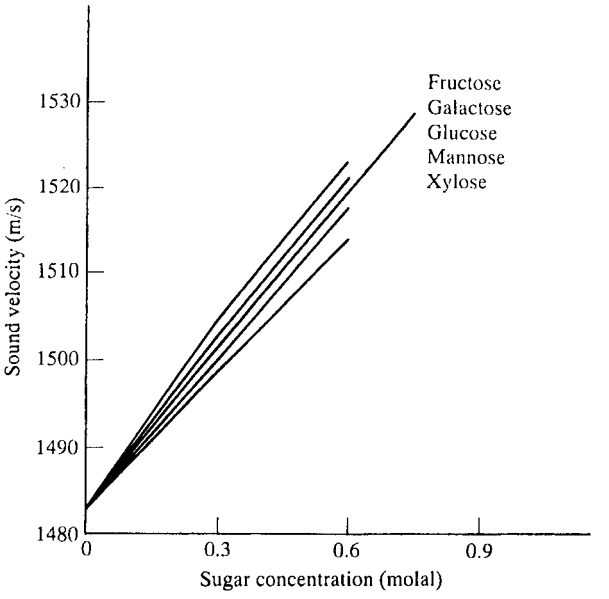


Fig. 13.6 Sound velocity as function of sugar concentration in aqueous solution for five monosaccharides at 20°C (Smith and Winder 1983).

The values obtained by Greenwood *et al.* (1999) with a prototype sensor for process control are consistent with the data shown in Fig. 13.7 (taking into account the temperature difference and the higher accuracy of laboratory measurements). The non-contact application of ultrasonic time-of-flight measurement to concentration measurement is attractive owing to its inherent hygiene advantage and protection from fouling by the sample. It is, however, restricted to those solutions either where only one of the relevant sample properties (solute composition, solute concentration, sample temperature) varies; or where variations in the other two can be monitored and compensated for. (The latter may be complicated by the change in the dependence on solute type with temperature.)

Qualitative monitoring of liquids by ultrasound velocity measurement is more easily implemented. An example is the distinction between a food liquid, a caustic solution (used periodically for cleaning-in-place or CIP) and plain water (used for flushing after cleaning). Control systems using ultrasonic instrumentation for this distinction have already saved many hectolitres of beer from entering the waste pipe.

When monitoring the mixing of two liquids, of which one or both are associated (such as water or alcohol), the non-monotonic dependence of the speed of sound on the mixing ratio (over part of the range) must be taken into account. For low concentrations of ethyl alcohol in water (less than about 20 per cent, depending on temperature), velocity increases with alcohol content; for high concentrations (above about 30 per cent), velocity decreases with increasing alcohol content (Blitz 1963). Similar observations have been made for another binary solvent mixture. At 30 Mol% of the second solvent, the speed of sound was 230 m/s higher than that for water whereas at 100% of the second solvent, the value was 10 m/s lower than for pure water (Syal *et al.* 1998).

For low alcohol concentrations in aqueous sucrose solutions, there is a temperature for which the velocity remains constant during yeast fermentation as ethanol is formed and

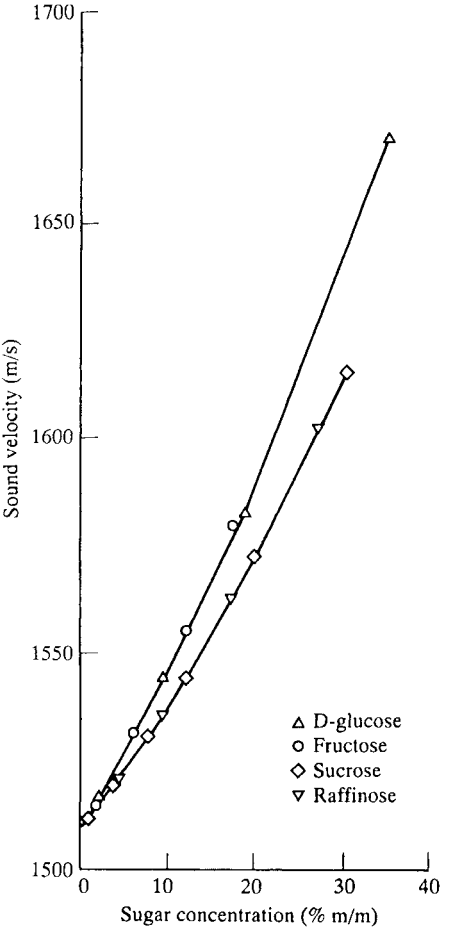


Fig. 13.7 Sound velocity as a function of sugar concentration in aqeuous solution for two monosaccharides and two higher saccharides at 30ºC (Schaafs 1963).

sucrose content decreases. The velocity at this temperature can be used as an indicator for the original gravity. Unfortunately, the temperature at which this occurs is above the process stream temperature for beer fermentation. For the determination of original gravity at process temperatures, a combination of sound velocity and refractive index measurement has been proposed instead. This has been tested successfully in the laboratory (Forrest 1987). In-line application would be more difficult in the presence of entrained gas or suspended yeast particles.

The ultrasonic properties of aqueous solutions of glycine, alanine and histidine have been examined by Wang and Feng (1990). In neutral solutions, the amino acid molecules exist in dipolar form and interact strongly with the surrounding water molecules to form a hydration area resulting in an increment to the sound velocity in pure water. For other pH values, a smaller velocity increment is observed. The hydration of polyamines in aqueous solution and its influence on the sound velocity have been investigated by Juszkiewicz and Figlerowicz (1990). Such studies have contributed to an understanding of both the ultrasonic properties and the molecular kinetics of biological media (as they are encountered in both clinical and food samples).

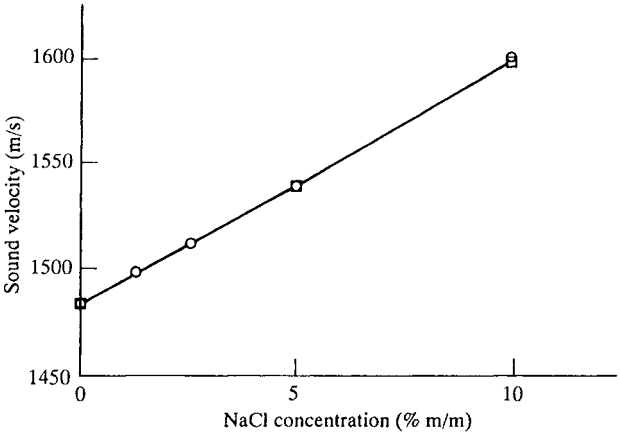


Fig. 13.8 Sound velocity as a function of concentration for aqueous NaCl solutions: × at 21°C and 1 MHz (Kress-Rogers 1986a); • at 20°C and 2 MHz (McClements and Fairley 1991).

It has been suggested that thick egg white could be distinguished from thin egg white by a measurement of ultrasound velocity or attenuation. As eggs are stored, thin white becomes more prevalent as the ovomucin content decreases. These eggs are then less suitable for the manufacture of certain food products. It would therefore be helpful if they could be distinguished from fresh eggs without being broken open. A method for the measurement of attenuation of ultrasound transmitted through whole eggs has indeed been presented (Section 13.3.5). However, literature data on the effect of ageing on the ultrasound characteristics of egg albumen are divergent (Povey and Wilkinson 1980; Javanaud *et al.* 1984; Choi *et al.* 1987).

The in-line measurement of ultrasound velocity in a foodstuff in order to derive its temperature has been proposed. This would be restricted to those applications where the process stream is predictable and constant in its other properties influencing the velocity. For liquid foods, this refers mainly to the precise chemical composition (including, for example, types and concentrations of dissolved saccharides or salts) but see also pages 374–80 on suspended solids or bubbles.

An alternative less liable to such cross-sensitivities is the use of a specially designed rod coupled to an ultrasonic transducer (see for example Gopalsami and Raptis 1983). The temperature of the rod material is then determined by measuring the speed of sound in that material. A particular section of the rod can be picked out for the temperature measurement by marker notches reflecting the signal. This rod section is immersed in the process stream. The method has been used in other industries, particularly in hostile environments, and has also been proposed for thermometry in the food industry (Povey 1989). It has been suggested that this method should be applied to semi-solid foods in microwave ovens. For plain water as the sample, this was shown to provide an accurate temperature measurement from approximately 30 to 55 °C (Richardson and Povey 1990).

For applications in semi-solid foods, however, such a probe would share a number of characteristics with other contact thermometers. The temperature of the food portion in good thermal contact with the probe tip would be measured. Temperatures in the remainder of the product would not necessarily be adequately represented. This would be particularly relevant in microwave ovens, where uneven heating can result from inhomogeneity in the dielectric properties of the foodstuff, from the shape of the food

item and from the oven design. Potential advantages over the current fibre-optic thermometers for industrial microwave ovens require clarification.

Influence of dispersed particles, droplets and bubbles

Solid particles dispersed in a liquid will influence the velocity of sound propagating through the material. This is due to the velocity difference between the liquid and solid phases of a fat. For common frying oils, for example, the difference in velocity between different oils at room temperature is of the order of 1 per cent (Table 13.1). The net increase (compensated for the temperature change) on solidification is of the order of 20 per cent (see figures in Miles *et al.* 1985). For frying fats such as lard or dripping, less than half of the volume is taken up by the solid phase at 20 ºC. The velocity differences between such fats at room temperature are due to the variations in the solid/liquid fat ratio as much as to the composition of the fats.

This has been studied for the assessment of solid/liquid ratios in fats where fat crystal seeds are dispersed in the liquid fat phase (Miles *et al.* 1985). The authors conclude that this method can estimate the volume fraction of solid in a partially melted fat (see Section 13.5.3 on the effect of particle size). They determined the second-phase content, that is the proportion of solid fat suspended in the liquid fat, by a measurement of the ultrasound velocity at 2.5 MHz.

McClements and Povey (1987) showed that a reproducibility of 0.2 per cent can be achieved in the determination of solid fat mass fractions (up to 20 per cent) dispersed in a liquid fat. It is worth noting that such a high reproducibility is dependent on a predictable composition of the fat. However, given that the difference between liquid and solid fats is much greater than that between common edible fats used for spreading or frying, this should not cause an undue problem. McClements and Povey conclude that the technique is easily automated, cheap, accurate, rapid, nonintrusive and based on sound physical principles, and should therefore prove useful both for in-line quality control monitoring and for food research applications.

Both groups of workers used Urick's (1947) equation, which describes the velocity as a function of the volume averages of compressibility and density. The model assumes that the dispersed particles are rigid, negligibly small compared with the wavelength, and present as a low mass fraction. Deviations from this model due to scattering are discussed in Section 13.5.3. Another application of ultrasonic velocity measurement suggested by McClements and Povey (1987) is the determination of the solubility of triglycerides.

Applications for ultrasonic instrumentation for the oils and fat sector before 1980 were largely based on empirical relationships. A better understanding of the underlying principles, together with advances in electronics, now provides the basis for wider and more reliable applications. A commercial system, once developed, would compete with the currently most widely used technique for solid/liquid fat ratio determination, namely pulsed nuclear magnetic resonance (NMR) (Sleeter 1985). Equipment cost and response time characteristics could be in favour of the ultrasonic method, but problems might be encountered with aerated samples such as some low-fat spreads.

Creaming profiles in oil-in-water emulsions can be obtained by relating the ultrasound velocity to the dispersed volume fraction at varying depths (Hibbard *et al.* 1986). In assessing a depth profile of creaming rather than trying to measure an absolute fat content, the technique is less sensitive to variations in dissolved and dispersed air content, the variable composition of the carrier liquid, and other factors influencing the velocity of ultrasound propagation.

Table 13.1 Sound velocity (measured) and impedance (calculated)
(a) Solids: machinery, container and transducer materials; biological tissues; solid foods

Material	Source of velocity	Velocity* c (m/s^{-1})	at T (°C)	Z(20°C) calc. † (10^6 kg m^{-2} s^{-1})
Steel, stainless	K	5980	~20	46.9
Steel, stainless	R	5790	~20	45.7
Quartz, ×-cut	K	5720	~20	15.1
Glass, Pyrex	K	5640	~20	13.1
Pine, 0% moist	SM	5080		
Pine, 50% moist	SM	4170		
Lead zirconate titanate (PZT)	C	~4000	~20	30
Bony tissue	SG	3365	16	6.12
Bone	C			8#
	K	4000	~20	
Sugar	P			4
Ice (high density)	K	3840	−20	
Tin, rolled	R	3320	~20	24.2
Polythene	R	1950	~20	1.76
Milk chocolate:				
deaerated	AA	2050	20	
untreated	P	≤1750	25	
aerated	P	≥900	25	
Cardiac muscle				
normal	O	1625		
(degenerate)		(1560)		

Table 13.1 (contd.)

Material	Source of velocity	Velocity* c (m/s^{-1})	at T (ºC)	Z (20ºC) calc. † (10^6 kg m^{-2}s^{-1})
Muscle, clinical	C			1.7#
Muscle:				
beef	SG	1572	18	1.63
veal	SG	1584	18	1.62
pork	SG	1589	18	1.63
mutton	SG	1588	18	1.64
Fish flesh	P	1550	22	1.6
Offal:				
liver	SG	1545	18	1.65
kidney	SG	1562	16	1.62
spleen	SG	1512	16	1.61
brains	SG	1503	16	1.52
Adipose tissue				
beef	SG	1431	18	1.37
pork	SG	1452	18	1.37
Cheeses	P	1360–1650	10	
Rubber, gum‡	R	1550	∼20	1.47
Teflon	AC	1340	∼20	2.95
Potato flesh	P	∼800	21	
Biscuits	PH	140–320	20	

Table 13.1 (contd.)

(b) Liquid and semi-solid foods: fats and oils; solutions, gels, emissions

Material	Source of velocity	Velocity* c (m/s^{-1})	at T (ºC)	Temp. coeff. dc/dT (m/s^{-1}K^{-1})	c(20ºC) (m/s^{-1})	Z (20ºC). calc.† (10^6 kg m^{-2}s^{-1})
Dripping	MF	1570			1570	
Lard	MF	1517			1517	
Milk fat	MS	1500	25			
Cocoa butter	P					1.5
Corn oil	MF	1477			1477	
Sunflower oil	McP	1470	20		1470	1.33
Peanut oil	Z	1450	25	−3.3	1467	1.33
Olive oil	Z	1448	25	−3.4	1465	1.32
	MF				1464	
Light corn syrup	Z	1958	25	−1.2	1964	
NaCl solution, 10%	A	1600	21		1598	1.71
NaCl solution, 5%	A	1539	21		1537	1.59
Sucrose solution, 50%	§	1740	10		1753	
Sucrose solution, 25%	§	1560	10		1579	
Glucose solution, 35%	¶	1670	30		1655	1.90
Glucose solution, 19%	¶	1583	30		1563	1.67
Egg white	P	1560	20		1560	
Apple juice	Z	1544	25	+2.2	1533	
Salad cream	McP	1528	20		1528	
Skimmed milk	P	1522	28			
Egg yolk	P	1500	20		1500	

Table 13.1 (contd.)

(c) Simple liquids

Material	Source of velocity	Velocity* c (m/s^{-1})	at T (ºC)	Temp. coeff dc/dT (m/s^{-1}K^{-1})	c(20ºC) (m/s^{-1})	Z (20ºC). calc.† (10^6 kg m^{-2}s^{-1})
Glycerol, pure	R	1904	25	−2.2	1915	2.42
Glycerine	K	1920	25	−1.9	1930	
Water, dist.	K	1447.3	10			
		1482.3	20		1482	1.48
		1509.1	30			
		1542.6	50			
		1554.8	70			
		1550.5	90			
Acetic acid	K	1173	20			
Ethyl alcohol	K‖	1145	25	−3.3	1162	0.94
Methyl alcohol	K‖	1103	25	−3.3	1120	0.88

(d) Ambient gases

Material	Source of velocity	Velocity* c (m/s^{-1})	at T (ºC)	Temp. coeff dc/dT (m/s^{-1}K^{-1})	c(20ºC) (m/s^{-1})	Z(20ºC). calc.† (10^6 kg m^{-2}s^{-1})
Air, dry	R**	331.45	0	0.59	343	0.0004
Nitrogen	K	354.4	29			
	R	334	0	0.6	346	
Oxygen	K	332.2	30			
	R	316	0	0.56	327	
Carbon dioxide	K	280 ($v_h = 293$)	51			
	R	259	0	0.4	267	
Ethylene	K	327	20		327	
Water vapour	K	477.5	100			

Table 13.1 (contd.)

* Velocity values are for compressional (longitudinal) waves in bulk material. Gases are at standard pressure unless stated otherwise.
Velocities will vary with the composition and processing history of the material. Values for metal alloys, glass, plastics or foodstuffs can also be regarded as typical only. Furthermore, some of the data were obtained many years ago with less sophisticated equipment. This also applies to some of the data cited in recent handbook editions or reviews.
The velocity at 20ºC is calculated, where necessary, with the help of the temperature coefficient or further velocity data in the source.
v_h is the high-frequency limit.
† Impedance Z calculated as $c(20ºC) \times \rho(20ºC)$ (unless marked #). Velocity values from sources as indicated; density values largely from *R*, one from C.
‡ Not the chewable variety!
§ Sucrose solutions: data for 10ºC interpolated for 50 and 25 per cent concentration from Figure 13.3c. Temperature dependence from F.
¶ Glucose solutions: velocity data for 30ºC from S, temperature dependence from F.
|| Ethyl and methyl alcohol: similar values also from G, but velocity value for ethanol is given as 1207 m/s^{-1} at 25ºC in R. Further values are cited in AC.
** Air: dry, 0.03 per cent CO_2, 1ºC, 1 atmosphere, 1 kHz: 331.45 m/s^{-1}. Adding water vapour: c first decreases (dispersion due to molecular relaxation), then increases (molecular weight (specific heats ratio)) (see K).

Sources:

A	Author, data for NaCl at 21ºC, 1 MHz (see Figure 13.6). Part of the data was published in Kress-Rogers 1986a. Temperature dependence for NaCl solutions derived from data in ICT 1930 combined with own data.
AA	Author, data for chocolate at 0.8 MHz. Related data were published in Kress-Rogers 1986b.
AC	Anson and Chivers 1990.
C	Crecraft 1983
F	Fedotkin *et al*. 1976.
G	Gray 1963.
K	Kaye and Laby 1986.
McF	McClements and Fairley 1991.
McP	McClements *et al*. 1990; salad cream values for typical UK formulation at 2.25 MHz.
MF	Miles *et al*. 1985.
MS	Miles *et al*. 1990.
O	Okawai *et al*. 1990.
P	Povey 1989.
PH	Povey and Harden 1981.
R	CRC Handbook 1976–77.
S	Schaafs 1963.
SG	Sajas and Gorbatow 1978.
SM	Sakai *et al*. 1990.
Z	Zaccharias and Parnell 1972.

For air bubbles dispersed in water, resonant scattering occurs when the frequency of measurement matches the resonant frequency of the bubbles. Lynnworth (1989) calculates the resonant frequency as 5.5 kHz/*d* (mm), for example 550 kHz for bubbles with a diameter of 10 m or 55 kHz for 100 m. The velocity of ultrasound propagation is that of water without air for frequencies well above resonance, but is reduced for lower frequencies. Lynnworth (1989) quoted a value of as little as 22 m/s (for 50 per cent air volume fraction, with a bubble size of 100 μm at 1 kHz). Thuraisingham 1998 has calculated the dependence of sound speed in bubbly water as a function of bubble size, bubble volume fraction and measurement frequency in the range 1 to 10 MHz. According to Thuraisingham's calculations, the speed of sound can reach values well above that for water, for example, 3000 m/s for bubbles of 10 μm in diameter with a volume fraction of 1 per cent measured at 4.5 MHz. For 50 μm bubbles MHz (also at 1 per cent volume fraction), the sound speed peak shifts to less than 1 MHz. For higher volume fractions, the resonance shifts to higher frequencies. The precise shape of the resonance curves might yet be modified by further calculations and experiments. However, when setting up process monitoring for a new type of sample, it would seem advisable to check for resonance effects by carrying out an experiment with varying ultrasound frequency if the sample might contain bubbles with a diameter of, say, 60 μm at a volume fraction of $\geq$1% and at a frequency of $\leq$3 MHz.

Attenuation of ultrasound is strongly enhanced for frequencies near resonance. This phenomenon can interfere with measurements, but can also help to assess the volume fraction and size of entrained gas bubbles in liquids (see Section 13.5.3). It is also used in drying processes (see Section 13.2.7).

Comparison of methods for composition monitoring

For the determination of water content and of other compositional variables, methods based on the physical interaction of the sample with ultrasonic, microwave and infra-red waves are discussed in this book. It is therefore appropriate to compare the characteristics of these three approaches and consider their suitability for particular applications in food process control. All three are also applied to the measurement of food properties other than composition. The comparison here is for composition measurement, and particularly water content determination, which is a common application for all three methods.

The in-line measurement of the water content of a solid or semi-solid sample is usually referred to as moisture content, whereas the water content of liquids would often be expressed as the complementary (dissolved or dispersed) solids content. This differs from the terminology in the analytical laboratory, where moisture can imply both water and other volatiles driven off in an oven-drying procedure.

Near infra-red (NIR) composition measurement is derived from highly specific mid-infra-red chemical analysis, but deliberately shifted to the near infra-red range to reduce specificity. NIR thus allows the quantitative monitoring of groups of compounds such as fat, protein or starch as well as of water. This multi-component analysis is a valuable advantage of near infra-red monitoring compared with microwave moisture or ultrasound solids content determination. With suitable reference wavelength selection and appropriate calibration procedures, reliable operation has been demonstrated for many industrial food applications. These are described in Chapter 6.

A disadvantage of NIR analysis (NIRA) in food process control applications is the generally low penetration depth. This necessitates a predictable relationship between the surface composition and the composition in the bulk of the sample. Problems are encountered when the surface is dried and hardened or covered with an oil film, for example.

Microwave methods offer an alternative when a bulk measurement of moisture content is required. This is then insensitive to variations in the distribution of moisture within the sample. Ultrasound is rarely applicable to the in-line monitoring of the composition of solid foods, but is applied to the determination of solute concentration. The following discussion centres on a comparison of water content measurement with microwave and ultrasound methods.

Microwave moisture measurement is characterized by the very distinct dielectric properties of free water (see Chapter 9). In contrast, ultrasound velocity in liquids is a function of compressibility and density. It depends quite sensitively on the composition of solids and liquids dissolved in the water. Accurate solute concentration measurement is thus confined to applications where the composition of the dissolved solids and the liquid are stable or defined by other simultaneous measurements.

Microwaves in the gigahertz range are strongly attenuated by water. They can penetrate only short distances into water or high-moisture materials, but they can readily be transmitted through deep layers of low-moisture materials. As the term 'microwave moisture' (as opposed to dissolved solids content) measurement implies, microwave techniques are used primarily for the determination of water content in solids, e.g. margarine, grains and powders. In such applications, microwaves can be transmitted through product streams travelling on conveyor belts or in pipelines. They will then provide a non-contact measurement sampling the entire process stream volume, yielding a bulk moisture value. Ultrasound cannot be applied to such solids-air systems owing to the high acoustic mismatch between the particulates and the air gaps. (NIRA is applicable provided that the surface is representative for the sample average.)

For the determination of solids content in aqueous solution, both microwaves (with stripline applicators) and ultrasound can be considered (as can NIRA). For homogeneous liquids with a low air content, ultrasound can often be transmitted through a substantial sample depth. For aerated liquids, on the other hand, ultrasonic time-of-flight techniques suffer from high signal attenuation. Acoustic impedance measurement can be considered but this is a measurement of the properties at the interface only and thus requires a representative sample layer there. Microwave measurements with stripline applicators can be used for both aerated and other liquids. For high water content combined with little aeration, they will have low penetration depth (but generally considerably deeper than with NIR reflection analysis).

A further aspect to be considered in the choice of a process control method is the windowing arrangement. Ultrasonic waves can be transmitted through a metal wall, although this can be frustrated by a high attenuation in the liquid inside the container, when most of the detected signal may have travelled around the vessel wall rather than through the liquid. For microwaves, free access over a conveyor belt, transmission through a Teflon window or the introduction of a thin stripline applicator can be considered. For near infra-red applications, free access or transmission through Aclar (PTFE polymer) or polyester/glass laminate shatterproof windows is used (see Chapter 6). Glass windows are also used, but the possibility of glass fragments entering the food process stream must be eliminated. For remote infra-red thermometry above 50 °C, sapphire windows have been suggested to provide the non-toxicity, non-solubility and strength that are important for safe applications in food process control. Fibre-optic applications have also been considered (see Chapter 7).

13.3.5 Speed of sound in solids

Of the applications of ultrasound measurement to the determination of the composition of solid food, the measurement of the ratio of fat to lean meat on carcasses is one of the most widely accepted. Ultrasonic pulse-echo reflection analysis is used in the assessment of back fat in live agricultural animals using paraffin as a couplant (Miles *et al.* 1983; Miles *et al.* 1984; Alliston *et al.* 1982). Further studies by Miles and coworkers have investigated the assessment of the proportion of ice in partially frozen food such as beef carcasses by ultrasound velocity measurement (Miles and Cutting 1974). In other sectors of the food industry, applications to a number of solid foods have been explored but are much further from industrial application. Techniques for the measurement of solid/liquid fat ratios in semi-solid fats are well advanced (see Section 13.3.4).

The characterization of solid foods such as fruit, vegetables (to assess conditioning or ripeness) and biscuits (to assess crispness) by ultrasonics has been studied. Industrial implementation would be complicated by the problem of providing good acoustic coupling without damage to objects passing at a high throughput rate, and by the high attenuation of some of these materials for ultrasound. There are also problems in interpreting the ultrasound characteristics of such cellular structures with models that work for steel or quartz. Crispness, together with other texture attributes, forms a complex food property that is difficult to correlate with instrumental measurements (see Chapter 18).

In laboratory studies, immersion tanks have been used to provide acoustic coupling. Under these conditions, slices of potatoes, for example, changed their properties, probably as a result of water absorption (Sarkar and Wolfe 1983). Povey (1989) reported ultrasonic measurements at 0.5 MHz on a wide range of fruit and vegetables. Slices of the produce rather than whole fruit had to be used to obtain a transmitted signal, so that a non-destructive measurement does not, as yet, appear to be feasible.

Soft-tipped probes, held onto the sample by spring-loaded clamps, have been used for biscuits to obtain ultrasound data (Povey and Harden 1981). Such probes have also been proposed for whole eggs (Povey and Wilkinson 1980) to measure both the egg albumen properties (see Section 13.3.4) and the egg shell thickness (Gould 1972; Volsey and Hamilton 1976). (Break strength would be of interest but cannot be obtained with low-intensity ultrasonic measurements.) For such fragile solid foods, it is difficult to establish a reliable measurement technique, a dependable correlation of ultrasonic data to industrially important food properties. Even more challenging is the implementation of such techniques in high-speed lines.

However, some progress has been made in this respect with the development of techniques for the measurement of acoustic properties at a distance (Sections 13.2.5, 13.2.4 and pages 391–2). One of these techniques is laser ultrasonics. This has been tested in the non-contact monitoring of the elastic properties (Young's modulus) of paper moving on a belt at 12 m/s (Ridgway *et al*. 1999). A potential drawback of this method is the localized increase in temperature which might affect delicate materials.

For cheese, ultrasonic monitoring of maturation has been proposed. Povey (1989) points out that this would require a measurement indicative of rigidity and breaking strength. The rigidity of cheese is low compared with its non-compressibility and is therefore difficult to measure with the proposed technique. The breaking strength is not accessible with low-intensity ultrasound measurements.

For food gels, the compressional wave velocity is usually indistinguishable from that in the equivalent sols prior to setting. The shear wave velocity is quite small (Section 13.3.2). In the measurement of speed of sound, the gel thus behaves like a liquid rather

than a solid. Attenuation increases on setting for certain gels (see Section 13.5.2). Progress has been made in non-destructive evaluation of biological tissues (Section 13.2.4) and this may find applications in the assessment of the microstructure and texture of foods.

13.4 Acoustic impedance

13.4.1 Definition

When an ultrasonic wave meets the interface between two media, the reflection and transmission rates are dependent on the ratio of the acoustic impedance of the two media. The acoustic impedance, in analogy to the impedance of a dielectric medium for electromagnetic waves such as microwaves (see Chapter 9), is governed by the ratio

$$\text{(excess pressure)/(particle displacement)} = E_{\mathrm{M}}k$$

The characteristic impedance is given by

$$Z = E_{\mathrm{m}}k/\omega = E_{\mathrm{M}}/c = c\rho = (E_{\mathrm{M}}\rho)^{1/2}$$

where again $E_{\mathrm{M}} = \gamma\rho$ for gases and $E_{\mathrm{M}} = K_{\mathrm{s}}$ for liquids. This is analogous to the impedance for the imposition of a longitudinal wave on bead-loaded coupled springs, where $Z_{\mathrm{sp}} = (K_{\mathrm{sp}}a\rho_0)^{1/2} = c_{\mathrm{sp}}\rho_0$ (with symbols as given on page xxvii). Z_{sp} is the constant of proportionality in the response (that is the force opposing the motion) of the spring to the imposed amplitude variation with time.

The specific acoustic impedance is usually given in the form

$$Z = \rho c$$

If follows from this relationship that the acoustic impedance will be smallest for gases (having both lower velocities c and much lower densities ρ at standard temperature and pressure (STP) compared with liquids) and highest for solids (see Table 13.1).

13.4.2 Implications for velocity measurement

When two adjacent media have impedances of the same order of magnitude, there is good coupling. This is the case for the interface between two metals or the interface between a soft solid such as nylon and a liquid. Impedances differing by less than two orders of magnitude are still reasonably well matched. This generally applies for a metal immersed in a liquid or a metal coupled to a less rigid solid such as Perspex.

The rate of transmission across a boundary is calculated from the coefficient of transmission

$$a_{\mathrm{T}} = 4Z_1Z_2/(Z_1 + Z_2)^2$$

For a solid–liquid interface with a ratio Z_1/Z_2 of the order of 0.1, the coefficient is about 0.3, leading to a loss of approximately 5 dB. For a solid–gas interface with a ratio of the order of 10^{-5}, the coefficient is about 4×10^{-5} and the loss amounts to about 44 dB. For a liquid–gas interface with impedances differing by four orders of magnitude, the loss would be about 34 dB.

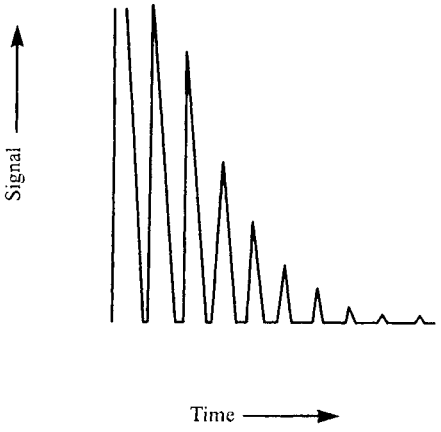


Fig. 13.9 Ultrasound pulse-echo pattern in a slice of redcurrant jelly (at ~20°C, 1 MHz; measured speed of sound $c = 1.6\,\mathrm{km\,s^{-1}}$).

For a liquid or semi-solid food sample (with modest attenuation for ultrasound) confined between solid parallel walls, the impedance mismatch at the two food–wall interfaces gives rise to an echo pattern from an ultrasonic pulse reverberating within the food sample and losing a proportion of the intensity at each reflection (Fig. 13.9). The distance between the pulse echoes can be used to measure the velocity of propagation within the food. The amplitudes in a multiple echo pattern can help to estimate attenuation in the food sample.

Only a small fraction of the acoustic energy is transmitted into a gas at the interface to a solid or a liquid. It is therefore necessary to fill even the minute air gap at the interface between two solids (resulting from the surface roughness) with a couplant. Immersion of the sample in water or the application of gels and greases on the sample surface are often employed in laboratory measurements to avoid air gaps. For long-term mounting, the transducer can be cemented onto a solid sample or container. The requirement for good acoustic contact (either directly with the sample or via a container wall in good contact with the sample) is a major hindrance for the application of ultrasonic techniques to in-line measurements on solid foods (see Section 13.3.5).

13.4.3 Measuring impedance instead of velocity

The measurement of the acoustic impedance of a sample can be used in place of a time-of-flight technique when the attenuation of the sample is too high for transmission. Clearly, this requires that the sample interface with the probe is representative for the bulk of the sample. In some applications, it will therefore be necessary to arrange for continuous wiping of the probe face. Reproducible coupling at the interface is also required for quantitative measurements of the sample impedance, and this will generally restrict the applications to liquids or samples that solidify before or during the course of the measurement without detaching from the probe. Further, temperature compensation will be required for such quantitative measurements just as in the case of time-of-flight measurements. Stable coupling only is required when the time dependence of a transition process is monitored, for example melting or setting behaviour or tempering characteristics in fat crystallization.

Randall *et al.* (1987) have used an impedance mismatch probe consisting of a buffer rod coupled to an ultrasonic transducer. The buffer rod, which is partly immersed in a

liquid food sample, ensures the resolution of the reflected pulse from the transmitted pulse so that the ratio of their amplitudes can be monitored. The transducer itself remains outside the process vessel. Buffer waveguides are also an option for hostile environments (Liu *et al*. 1998).

A similar arrangement is used by McClements and Fairley (1991) as part of an instrument for the determination of ultrasonic velocity, attenuation coefficient and characteristic impedance of a liquid sample. This 'ultrasonic pulse-echo reflectometer' is intended for adaptation to the automatic determination of a number of physical properties of liquid samples in tanks or pipes. Most readily determined is the density as $\rho = Z/c$. For the monitoring of the concentration of dissolved solids, impedance Z can be measured as an alternative to velocity by time of flight for materials with high attenuation. As in the case of direct velocity measurement, the calibration for the application in question needs to be established first, and variations in solute or solvent composition or in the temperature will affect the readings.

Acoustic impedance measurements can also be carried out with the help of Lamb wave devices (Sections 13.2.5 and 13.3.2). The acoustic impedance mismatch at the interfaces between a liquid and solids carried by the liquid can be exploited for the ultrasonic tomography of such process liquids (Section 13.2.3).

13.5 Attenuation

13.5.1 Sources of attenuation

The attenuation of ultrasound in gases is relatively high and increases with frequency. Also, their acoustic impedance is low compared with that for solids so that only a small fraction of the energy from the transducer is transmitted into the medium. Measurements at frequencies in the megahertz range are therefore impractical for gases. Classical theory predicts an increase of the coefficient of sound attenuation with the square of frequency, and this holds for most common gases (but not for carbon dioxide; see the next section) in the ultrasonic range for process control measurements (up to 100 kHz). This frequency dependence also applies to most common liquids within the ultrasonic range considered here (up to about 10 MHz), but the attenuation is often much higher than predicted by classical theory, even for pure liquids without entrained bubbles or suspended solids.

Attenuation can arise from

- a relaxational type of absorption, or
- deviation of energy from the parallel beam.

In the first case, an energy mode with a time lag to establish equilibrium conditions (for example, thermal conduction, viscous flow, vibrational motion of molecules) is coupled to an energy mode with instantaneous transfer (such as translational motion). With increasing frequency, the phase lag increases (thus increasing absorption) but the probability of energy transfer decreases (thus decreasing absorption). As a result, a broad maximum in the loss per cycle is observed at the relaxation frequency ω_0. In the second case, the beam is scattered or diffracted by inclusions of a differing acoustic impedance, or reflected and refracted at interfaces (boundaries) between different acoustic media.

For monatomic gases, absorption is due to viscosity and thermal conduction. Both can be described as relaxational phenomena, but with relaxation frequencies well above the ultrasonic range for industrial process control measurements. Within this range, absorption is therefore approximated by classical theory. The loss per cycle is

proportional to the frequency, the coefficient of absorption increases with the square of the frequency, and the velocity of propagation is independent of frequency. For diatomic gases at STP, the classical description is still adequate within the ultrasonic range considered here. Considerable discrepancies can be observed for polyatomic gases (see Section 13.5.2).

For most liquids, absorption is dominated by viscous and vibrational relaxation, and for associated liquids such as water or alcohol also by structural relaxation. Only the contribution from viscous relaxation is included in the classical description. This accounts for the attenuation of associated liquids well in excess of the classical description. For rotational isomers, for example the ester ethyl acetate, a relaxation frequency just within the ultrasonic range considered here can also occur. Food liquids often contain entrained gas bubbles or suspended solid particles, and these give rise to high attenuation (see Section 13.5.3).

13.5.2 Relaxation

For certain media, an unusually high attenuation not following the frequency dependence expected from classical absorption theory can be observed within a part of the ultrasound range. This is, for example, the case for carbon dioxide, where the loss per cycle exhibits a broad maximum within the ultrasound frequency range used for transmission through gases (shifting to higher frequencies for carbon dioxide with water vapour inclusions).

Similar characteristics are observed for other gases composed of linear triatomic molecules. This broad peak in attenuation at a characteristic frequency is due to thermal relaxation associated with the exchange of acoustic energy with an internal vibrational mode of the CO_2 molecule. It is accompanied by velocity dispersion, that is the velocity is now frequency dependent. A practical consequence is the restriction of ranging applications (for level measurement) to headspace gases with low carbon dioxide content (see Section 12.2.1).

While the high attenuation of carbon dioxide due to relaxation effects is a nuisance in ranging applications through a CO_2-filled headspace, it may be the basis of a possible application in the determination of the concentration of other gases admixed to CO_2. This is relevant in packaging under a protective carbon dioxide environment to suppress microbial activity. A small proportion of oxygen is added to the carbon dioxide when the activity of anaerobes (such as *Cl. botulinum*) must be prevented. Ultrasonic attenuation is highly sensitive to small changes in the concentration of a non-relaxing gas admixed to CO_2.

For attenuation due to relaxation, characterized by a relaxation time τ, and for a dispersive sound velocity $v(f)$ with a low-frequency limit c, the coefficient of attenuation is

$$\alpha = \frac{\omega^2 \tau v}{2c^2(1+\omega^2\tau^2)} = \frac{v/\tau}{2c^2}\frac{\omega^2\tau^2}{1+\omega^2\tau^2}$$

The loss per cycle is

$$Q^{-1} = \frac{\alpha\lambda}{\pi} = \frac{2\alpha}{k} = \frac{v^2}{c^2}\frac{\omega\tau}{1+\omega^2\tau^2}$$

where ω is the angular frequency $(2\pi f)$, and a broad maximum of Q^{-1} occurs at $\omega = 1/\tau = \omega_0$.

For frequencies well below the relaxation frequency, that is for $\omega \ll \omega_0 = \tau^{-1}$, these equations are simplified to

$$\alpha_{lf} = \frac{\upsilon^2}{2c^2\tau}\omega^2\tau^2 = \frac{\upsilon\tau}{2c^2}\omega^2 = \frac{\tau\omega^2}{2c^2}$$

$$Q_{lf}^{-1} = \frac{\upsilon^2\tau}{c^2} = \tau\omega$$

The coefficient of attenuation now increases with f^2 and the loss per cycle is directly proportional to frequency f.

For most common gases, absorption is adequately described by the contributions from viscosity and thermal conduction, where

$$\tau_{vis} = \frac{4}{3}\frac{\eta}{\rho c^2}$$

$$\tau_{thc} = \frac{\kappa}{\rho c^2 c_\upsilon}\frac{\gamma - 1}{\gamma}$$

In the low-frequency limit $\omega \leq \omega_0$, the corresponding attenuation coefficients are

$$\alpha_{vis,lf} = \frac{8\pi^2}{3}\frac{\eta}{\rho c^3}f^2$$

$$\alpha_{thc,lf} = \frac{2\pi^2}{\rho c^3 c_\upsilon}\frac{\gamma - 1}{\gamma}f^2$$

The classical prediction is equal to the sum of these low-frequency limits of the relaxation model:

$$\alpha_{\text{class}} = \alpha_{\text{vis, lf}} + \alpha_{\text{thc, lf}}$$

The attenuation is inversely proportional to gas density, accounting for the difficulty of transmitting ultrasound through a gas at low pressure. Values for relaxation times in gases and liquids are given in Table 13.2a; they are put into the context of the ultrasonic spectrum in Fig. 13.10. Calculated classical absorption coefficients are given in Table 13.2b, and measured absorption coefficients in Table 13.2c.

Relaxation studies for gels have been described by Wyn-Jones *et al.* (1982) and by Ross-Murphy (1984). For random-coil polymers, a substantial attenuation was observed in the range 1–100 MHz. This feature was found to be less pronounced for conformationally ordered biopolymers. For the gels of the polysaccharide agarose, the increase in attenuation compared with the sol (prior to setting) was found to be a function of concentration. For 3 per cent w/v agarose in water, the attenuation coefficient at 1 MHz was by a factor of about 10 higher in the gel than in the sol. The difference decreased with increasing frequency (Wyn-Jones *et al.* 1982). Relaxation phenomena have already been described in Chapter 9 in the context of composition measurements based on the interaction of microwaves with the sample. For the propagation of electromagnetic waves in dielectric media, relaxation was shown to be characterized by a broad maximum in dielectric loss and a decrease in permittivity as frequency increases (Fig. 9.1). The absorption of microwaves by water in the relaxation regime is the basis of microwave cooking.

Table 13.2 Relaxation times and absorption coefficients
(a) Relaxation times (seconds) for gases and liquids at 20ºC and STP (Blitz 1963, pp. 87–105, 117–123)

Relaxation times*	Gas	Liquid
τ_{vis}	$\sim 10^{-10}$ 1.6×10^{-10} air	$\sim 10^{-12}$ 0.6×10^{-12} water 2×10^{-12} ethanol§ 60×10^{-12} olive oil 400×10^{-12} glycerol¶
τ_{thc}	$\sim 10^{-10}$ 0.7×10^{-10} air	$\sim 10^{-17}$
τ_{rot}	$\sim 10^{-9}$ N_2, O_2	
τ_{vib}†	$\sim 10^{-5} CO_2$ $\sim 10^{-7} SO_2$ $\sim 10^{-8} H_2O$ vapour	$\sim 10^{-10}$
τ_{str}		$\sim 10^{-10}$ glycerol $\sim 10^{-12}$ water $\sim 10^{-14}$ *n*-propyl alcohol
τ_{iso}		$\sim 10^{-7}$ ethyl acetate (ester)
$1/(2\pi f_{max})$‡	$\sim 10^{-6}$	$\sim 10^{-8}$

* vis viscosity
thc thermal conduction
rot molecular rotation
vib molecular vibration
str structural
iso rotational isomers
† τ_{vib} is the longest quoted if several present.
‡ f_{max} is the highest frequency used in the industrial monitoring applications described in Chapter 12.
§ Ethyl alcohol is the better label but less commonly used.
¶ Pure glycerol, known as glycerine in extracts of varying imurity.

(b) Calculated absorption coefficients for classical model α_{class}/f^2(Np m^{-1} Hz^{-2})*, at 20ºC and STP (Blitz 1963, pp. 88, 117)

Gas	Liquid
$\sim 10^{-11}$ 1.4×10^{-11} air 1.3×10^{-11} CO_2†	$\sim 10^{-14}$ 0.8×10^{-14} water 3×10^{-14} ethanol ‡ 80×10^{-14} olive oil 600×10^{-14} glycerol§

* Neper and decibel: 1 Np = 8.686 dB (where 8.686 = 20 lg e = 20/ln 10).
Expression of intensity ratios: 10 lg (I_2/I_1) dB or 0.5 ln (I_2/I_1) Np.
† For CO_2 the calculated attenuation is particularly unrealistic, as τ_{vib} is longer than the period of the measurement frequency.
‡ Ethyl alcohol is the better label but is less commonly used.
§ Pure glycerol, known as gltcerine in extracts of varying impurity.

(c) Measured absorption coefficients, $\alpha_{meas}/f^2 (\times 10^{-14}$ Np m^{-1} Hz$^{-2})$ (Kaye and Laby 1986; Gray 1963; Anson and Chivers 1990)

Substance*	At 25ºC	At 20ºC	
Water	2.2	2.6	
Ethanol†	5	9.0	
Glycerine‡	300		
Olive oil		135.0	
Steel (tool)		4.9	(10 MHz)
Quartz (x-cut)		0.013	(10 MHz)
Glass (crown)		2.0	(10 MHz)
Polythene		5 400	(1 MHz)
Rubber		12 000	(0.35 MHz)

* For solids, the measurement frequency is given in brackets.
† Ethyl alcohol is the better label but is less commonly used.
‡ Glycerol in extracts of varying impurity.

In the case of microwave moisture measurement, for example, the engineer testing a microwave attenuation monitor for a new group of materials may find that when the sample moisture rises beyond a certain value, the instrument output is no longer increasing gradually with the moisture content, but instead reaches a maximum and then falls. Having identified the relaxation characteristics, one can then try a more appropriate frequency range for the attenuation measurement (Fig. 9.3b), or switch to microwave phase shift (Fig. 9.3a) as the measured variable.

In the application of ultrasonic techniques in food liquids and headspace gases, relaxational phenomena can irritate the engineer with unexpectedly high attenuation levels for certain media. For high carbon dioxide contents as they occur in the headspace of fermenting or carbonated liquids, strong attenuation can preclude measurements involving ultrasound transmission through the headspace. For most food liquids, the

Gases	*Liquids*	*Frequency* f_o (Hz)
	τ_{str} water	10^{11}
	τ_{vis} water, ethanol	
		10^{10}
	τ_{str} glycerol	
τ_{thc} gas	τ_{vib} olive oil	10^9
τ_{vis} gas	τ_{vis} glycerol	
τ_{rot} N_2, O_2		10^8
τ_{vib} H_2O vapour		10^7 L
τ_{vib} SO_2	τ_{iso} ethyl acetate	10^6 L
		10^5 G
τ_{vib} CO_2		10^4

Figure 13.10 Relaxation frequencies within the sound spectrum (see Table 13.2a). G (gases), L (liquids indicate typical frequency ranges for the industrial process control measurement described in Chapter 12. Figure shows that for ultrasound transmission through CO_2 or ethyl acetate, severe attenuation and a dispersive propagation velocity have to be expected.

relaxation frequencies will be above the ultrasonic frequency used in process measurement (except perhaps in the case of esters). However, for many food liquids, the measured attenuation will be in excess of the values calculated from classical theory (even without the inclusion of scatterers). It is usually this increased attenuation, rather than the frequency dependence or the additional pressure and temperature dependence of the dispersive velocity, that is likely to present a problem in process measurement in the food industry. A detailed discussion of relaxation phenomena in ultrasonics can be found in Blitz (1963) and in Kinsler *et al.* (1982).

13.5.3 Scattering

Hindrance or help?

For liquids, attenuation at high frequencies can be very high where entrained air bubbles or suspended solid particles are present, as is often the case in food liquids and semi-solids. During food processing operations, pumping and mixing can lead to substantial air entrainment and heavy frothing, and this can spoil an ultrasound velocity measurement that was possible in the tranquil laboratory sample. Attenuation is also high for foods such as chocolate where the liquid fat phase carries not only suspended solid fat crystal seeds, but also solid sugar particles and air inclusions. The sugar particles have an acoustic impedance much higher than that of the cocoa butter (by a factor of 3; see Table 13.1(a), (b)) and are therefore a cause of considerable scattering.

In such two-phase (or three-phase) systems, a lower frequency for ultrasound transmission may be chosen to reduce attenuation (Kress-Rogers 1986b) where Rayleigh-type scattering (see later) is dominant. Frequencies of 0.5 to 0.8 MHz have been used for measurements on chocolate (see Table 13.1(a)). Alternatively, a different ultrasound technique (based on the measurement of acoustic impedance, for example; see Section 13.4.3) can be considered. A hindrance in some applications, the attenuation due to scattering is the basis of others, as described below.

Concentration and characterization of suspensions

The increase in attenuation with suspended solids inclusion is the basis of a technique used in the food industry for the determination of the solids content of slurries up to 50 per cent w/w (see Chapter 12, Section 3.4). Also, small air bubbles can be detected even at low volume fractions under resonant scattering conditions (Section 13.3.4). For concentrated suspensions such as foams or slurries with high dispersed solids contents, a full theoretical description is complicated by the complexity of the scattering properties of such systems (Povey and McClements 1988).

Applications in other industries have demonstrated that it is possible to detect dispersed air bubbles, oil droplets and solid particles and distinguish them from each other by ultrasound measurements (including attenuation, forward and backward scattering). This has proved applicable to high flow rates of the process stream. Examples include air (as bubbles, in foam or slugs) in blood during processing, oil droplets in recirculated water during secondary oil recovery, wear particles in engine oil, and small suspended air bubbles (as distinct from large loose bubbles) in paper pulp suspensions (Lynnworth 1989).

The ultrasonic attenuation and backscatter coefficient from suspensions are correlated with the concentration of particles and the particle size distribution (Groetsch 1990, ALBA Felix 1991). This method of suspension characterization is not suitable for very low particle concentrations (< 500 ppm). For such applications, an ultrasound reflection

technique has been developed. Using this method, the particle size distribution and concentration can be obtained for materials such as oil drops in water or undissolved particles in solutions (Volker and Kroon 1998).

Juice stability and product defects

The scattering of ultrasound by colloidal particles has been studied by Sarkar and Wolfe (1983) in monitoring the separation of reconstituted orange juice. They measured the attenuation of ultrasound at 1 MHz transmitted through the juice. The separation level of the settling colloidal particles could be observed in the attenuation as a function of liquid depth. This technique is intended for the non-destructive inspection of reconstituted juice in cartons.

Backscatter of ultrasound at 5 MHz has been studied by the same authors for characterizing the smoothness of orange skin, detecting cracks in tomato skin and distinguishing between normal and partially filled or shrivelled cobs of husked sweet corn (Dull 1986; Sarkar and Wolfe 1983). The produce was mounted in an immersion tank. (Optical sorters would probably be easier to apply, as long as there was a visible defect; see Chapter 5.) For the assessment of ripeness or conditioning of produce, attenuation of submerged slices was measured (see Section 13.3.5).

Flow metering and flow imaging

Scattering from suspended solid particles or droplets (of a second immiscible liquid) or of gas bubbles in a liquid is used in the Doppler flow meter (see Chapter 12, Section 13.3.1). This is not applicable to molten chocolate, unfortunately, despite the abundance of scatterers there. The attenuation here is too high and the signal, if any, would originate from the outer layer of the flow profile, possibly a stagnant part of the sample stream. Doppler flow meters can, however, be used for the monitoring of some components used in chocolate manufacture (McFarlane 1987).

Both transit-time and Doppler flow meters are based on a measurement of flow velocity. This has to be combined with an assumption on the flow profile and the part of this that is actually sampled by the flow meter in order to derive the mass flow rate. Food liquids often exhibit non-parabolic flow profiles and, unless their flow rate profile is known and taken into account, the flow rate reading will be unreliable. Time-gated Doppler flow meters can give information on the flow velocity profile. A new method for the estimation of the mean axial velocity in ultrasound Doppler flow measurement has been presented by Munk and Jensen (2000).

In addition to flow rate, the Doppler method can also provide information on the presence and size of obstructions when a special evaluation technique is used (Bastos *et al*. 2000). For larger particulates, the flow rate can differ from that of the carrier liquid. Also, their distribution across the flow profile can be uneven. For such sample streams, special care is needed in the interpretation of flow measurements. Bubbles can also have a flow rate differing from the carrier fluid. The air flow rate can be measured non-intrusively by laser ultrasonics (Horvat *et al*. 1998).

Ultrasonic flow meters based on the Doppler shift or on transit time are discussed for food applications in Chapter 12, Section 3.1, and for general applications by Lynnworth (1989, 1994). Lynnworth also describes vortex shedding flow meters, which are widely used in other industries. In the food industry, the presence of the bluff body in the process stream (required for this technique) is a disadvantage from the hygiene point of view.

To isolate ultrasonic transducers from the process in hostile environments, buffer waveguides have been developed (Liu *et al*. 1998). Another option for hostile

environments or for the rapid scanning of large areas in applications such as flow measurement and position sensing are systems measuring at a stand-off distance from the process. Ultrasonic electrostatic transduction methods for measurement in both air and water have been developed by Hutchins *et al*. (1998). (See also Sections 13.2.4 and 13.2.5 on stand-off measurements.)

Cross-correlation techniques (Beck and Plaskowski 1987) are a further non-invasive alternative. In cross-correlation flow metering, the transit times of random fluctuations in the scattered signal (its amplitude, phase and frequency) are measured. These can be due to turbulent eddies or to density fluctuations in two-phase liquids, for example.

As a special flow measurement application, mass flow measurement is described by Lynnworth (1994). In addition to the volume flow rate, liquid density is determined by measuring the ultrasonic propagation velocity for a flowing liquid of known temperature and composition. One of the suggested measuring arrangements is a pipe section of uniform wall thickness carrying one pair of ultrasonic transducers for the measurement of volume flow rate, a further pair for the measurement of ultrasonic velocity in the liquid and one temperature probe. It is used primarily in clinical applications for the determination of blood flow characteristics. Another special application of ultrasonic flow measurement is 'colour flow mapping'. The flow is mapped in a three-dimensional representation. This shows the distribution of flow rates within a flow cross-section (Lynnworth 1994; Rickey *et al*. 1991; Kasai 1993). The method has been extended to three-dimensional ultrasonic imaging (Fenster *et al*. 1998). On ultrasonic tomography for industrial applications, see Section 13.2.3.

Emulsions: the three contributions to scattering losses

Scattering is weaker for oil-in-water emulsions where a liquid phase is dispersed within a second liquid phase of comparable compressibility $1/K$ and density ρ. In an ideal non-scattering two-phase system, the velocity is calculated from the volume averages of the respective values of K and ρ for the component phases (the velocities themselves are not averaged); attenuation is calculated as a volume average. In real food emulsions, however, both the velocity and the attenuation depend on the average size and the size distribution of the droplets as well as on the volume proportion of the second liquid phase. This has been demonstrated for salad creams with ultrasound measurements at 1–6 MHz (McClements *et al.* 1990). The authors suggest a measurement at two frequencies for a simultaneous in-line determination of droplet sizes and fat content. Ultrasonic particle sizing offers the advantages of non-invasive *in-situ* measurement without prior dilution over particle sizers on the market now.

The excess attenuation (additional to the volume average) is attributed to three contributions: simple scattering (Rayleigh type in the long-wavelength limit); scattering due to viscous loss; and scattering due to thermal loss. The latter two contributions are included under the heading of 'scattering' by some authors; others reserve this term for simple scattering. For each of the three contributions, the increase in the coefficient of attenuation is proportional to the volume fraction of the suspended particles but depends also on other sample properties.

The three contributions have been discussed for a single particle within a medium in Landau and Lifschitz (1975). A theoretical description for emulsions and suspensions including the three contributions was later given by Allegra and Hawley (1972). The thermal properties of the fluid and suspended second phase have only recently been included in the prediction of ultrasound propagation in food emulsions. McClements and Povey (1989) have examined simple oil-in-water emulsions; McClements *et al.* (1990)

have studied salad creams (see above). Miles *et al.* (1990) have investigated milks and creams. In the following description, a non-dispersive velocity regime (see Section 13.5.2) is assumed, and this will apply in most food industry applications of ultrasound.

In the simplest analysis of ultrasound propagation in dilute suspensions, lossless spherical particles in a lossless inviscid fluid are taken to be the model. When the particle radius r_p is small compared with the wavelength, the coefficient of absorption increases with $(f/c)^4$, ${r_p}^3$ and with the difference in compressibility and density between the liquid and the suspended particles. For milks and creams at 2–7 MHz, this contribution is negligible.

The second contribution is due to the effect of the fluid viscosity on the motion of the particles. This relative motion must occur when the density of the particles differs from that of the carrier liquid. This contribution is a complex function of frequency, density ratio, particle radius and viscosity.

The third contribution is due to the conduction of heat between the fluid and the particles. The pressure variations travelling with the ultrasonic wave are accompanied by temperature variations, and the differing thermal properties lead to a temperature gradient at the particle surfaces and thus to heat flow between the components of the emulsion. For milks and creams at 27 MHz, this contribution is dominant and decreases with increasing particle size. The same was true for the salad creams at 1 MHz.

The relative importance of the thermal contribution to the attenuation in liquids containing suspended solid particles has been studied for a wide range of liquids by Anson and Chivers (1990). They conclude that the thermal contribution must be included in virtually all systems for $ka \leq 0.5$ (where $k = 2\pi/\lambda$). The relative importance of the other contributions depends critically on the values of the relevant parameters (see above) of the two phases. For $ka \geq 1$, the thermal contribution needs to be considered only for liquids containing a second liquid phase in suspension and where these two liquids are similar in all but the thermal properties. Otherwise, simple scattering and viscoinertial scattering dominate in this regime. For solids suspended in liquids, simple scattering dominates.

For milks, creams and salad creams (see above), the mean fat droplet size (about 0.5–10 μm diameter, depending on homogenization treatment) was much smaller than the wavelength at 1 MHz (about 1.5 mm). The excess attenuation was dominated by thermal conduction between fat globules and fluid phase. The simple, Rayleigh-type scattering contribution was negligible here and the attenuation increased with homogenization treatment, that is with decreasing particle size. For milks (with fat contents up to 4 per cent), attenuation increased in proportion with fat content for a given particle size. The measurement of both the degree of homogenization and of the fat content of milk is of industrial interest.

13.6 Conclusions

A range of applications of low-intensity ultrasound are well established for process control in the food industry. Liquid level, solids level and liquid flow as well as suspended solids concentration (for constant particle size) are routinely being determined. New applications are continually emerging based on developments in ultrasonic instrumentation and in the characterization of the ultrasonic properties of foodstuffs. Solute concentration is being monitored in well-defined applications. Further sample characteristics are being inferred, for example droplet size in oil-in-water emulsions, crystallization characteristics in fats or setting behaviour in gels.

The complex relationships between the propagation characteristics of ultrasound and the physical and chemical properties of foodstuffs and their environment are the basis of this wide range of application areas, but they are also a source of cross-sensitivity. With due care in the selection of new applications, the scope of industrial ultrasound techniques can be expanded to probe a range of food properties in addition to established process variables.

The simultaneous measurement of several variables is becoming more widely applicable as an option in process measurement with the advances in electronics employed in measurement and signal processing. For example, a combination of ultrasound velocity, impedance and/or attenuation can be used, or an in-line measurement of ultrasound velocity can be combined with that of refractive index (optical). Such multi-variable measurements provide a more reliable performance for many existing applications and allow new industrial applications to be introduced. New developments in transducing systems and signal evaluation are paving the way to ultrasonic scanners for industrial processes and to the characterization of the microstructure of soft solids.

The option of the non-contact monitoring of processes with ultrasonic methods has always been an attractive feature for the food industry where the processed materials should be in contact only with components that can be subjected to rigorous cleaning-in-place procedures. Additionally, ultrasonic measurements at a stand-off distance have now become possible in some applications due to the development of new ultrasonic transducer types and of laser ultrasonics. Process monitoring based on monitoring the acoustic emission generated by the process itself has been further refined.

13.7 References

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Appendix: Ultrasound measurement applications in and for the food industry

As this chapter describes the principles underlying ultrasound measurement in the food industry, the sequence here is determined by the measured ultrasound variable such as speed of sound or acoustic impedance. In Chapter 12, applications of ultrasonics currently established in the food industry are discussed and the sequence there is by instrumentation type.

In addition to giving an overview of ultrasound applications in and for the food industry, this appendix is intended to help locate the section describing the measurement of a particular target variable of interest to the reader, such as flow rate or solute concentration. The sections are keyed as follows:

12.*n.n* Application details for current instrumentation discussed in Chapter 12, Section *n.n.*

13.*n.n* Application, either implemented or at the research stage, discussed in Chapter 13, Section *n.n*, in the context of the underlying principle.

For data on the ultrasound properties of specific foods and literature sources, see Tables 13.1, 13.2 and Figs 13.3–13.6.

Measurement of process variables

Fill level indication:	
liquids	12.2.1, 12.3.2, 12.3.3
solids	12.2.1
Flow rate metering, liquids	12.3.1, pp. 391–2
Mass flow metering	pp. 391–2
Counting of items passing	12.2.2
Detection and sizing of flow obstructions	p. 391
Scan of pipe cross-section for liquid carrying solid items	13.2.3
Interface detection and location:	
liquid to liquid	12.3.4
emulsion to single phase	12.3.4
Classification of liquid content of passing containers	13.2.5
Characterization of sheet-type solid passing on moving belt	13.2.5

Measurement by acoustic emission monitoring

Powder flow	13.2.6
Cleaning fluid impact	13.2.6
Cooker extruder performance	13.2.6
Drying, grinding	13.2.6

Measurement of the properties of liquid foods

Distinction between:	
foam and liquid	12.3.2
cream and milk	12.3.2
beer and cleaning solution	13.3.4
Concentration of solutes and solvents in aqueous solution:	
NaOH (in cleaning solution)	12.3.5
NaCl	13.3.4, 12.3.5
monosaccharides (fructose, galactose, D-glucose, mannose, xylose)	13.3.4
disaccharide (sucrose)	13.3.4, 12.3.5
polysaccharide (raffinose)	13.3.4
coffee extract	12.3.5
modified corn starch	12.3.5
ethyl alcohol	13.3.4
Original gravity of beer before, during and after fermentation: measure sound velocity + optical refractive index	13.3.4
Hydration:	
amino acids	13.3.4
polyamines	13.3.4
Mass or volume fraction in liquid carrier:	
suspended solids (slurry)	13.5.3, 12.3.4
suspended solids (fat crystals in liquid fat)	13.3.4
dispersed droplets (in milk, cream, salad cream)	13.5.3
dispersed bubbles	13.3.4
Size and size distribution:	
small air bubbles in liquid	13.3.4
droplets in emulsion (milk, cream, salad cream)	13.5.3
solids suspended in liquid	13.5.3
Creaming:	
oil-in-water emulsions	13.3.4
Settling (separation, sedimentation):	
juice colloids	13.5.3
Temperature	13.3.4
Density	13.4.3

Note: see also Table 13.1b for references to specific liquid foods.

Measurement of the properties of solid and semi-solid foods and of animals

Back fat, live animals	13.3.5
Fat/lean ratio, carcasses	13.3.5
Tissue structure/texture	13.2.4
Frozen/liquid water ratio in meat	13.3.5
Solid/liquid ratio (fat)	13.3.4
Melting, solidification	13.4.3
Crystallization, tempering	13.4.3
Solubility (triglycerides)	13.3.4
Setting (gels)	13.4.3, 13.5.2
Temperature	13.3.4

Note: see also Tables 13.1a, b for references to specific solid and semi-solid foods.

Potentially more difficult measurement applications

Freshness of eggs (thick/thin egg white ratio)	13.3.4
Maturing of cheese	13.3.5
Crispness, biscuits	13.3.5
Ripeness, conditioning (produce)	13.3.5, 13.5.3
Smoothness, cracks, shrivelling, missing kernels (produce)	13.5.3
Thickness, egg shells	13.3.5

Note: see also Tables 13.1a, b for references to specific solid and semi-solid foods.

Applications not feasible

Determination of breaking strength	13.3.5
Echo ranging through air containing high carbon dioxide levels	13.5.2, 12.2.1